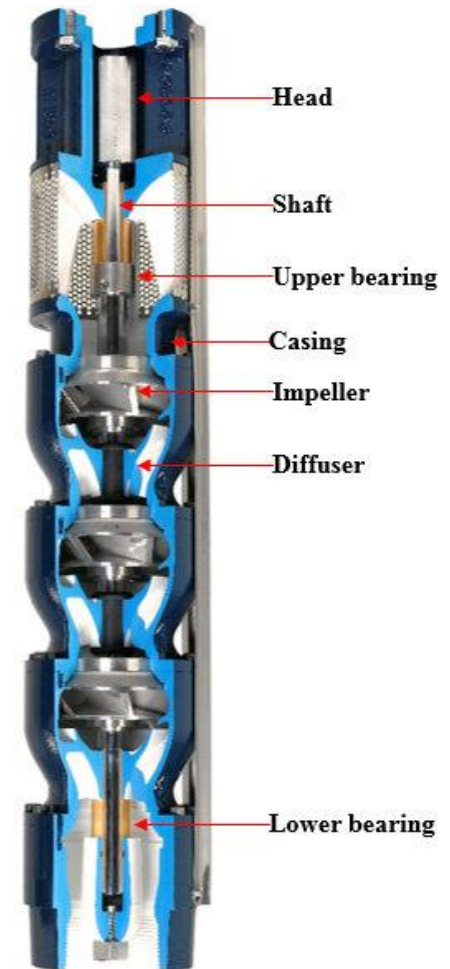


The E.S.P. Problems & Troubleshooting

Petroleum Engineering Lecture
Prof. Hussain Ahmed M.B.E.

Saturday 10/10/2020
5 pm. London time.

ZOOM ID: 823 9642 1478



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10.10.2020

ESP Overview

Agenda

- ESP Main parts
- ESP; How it works
- When do we need ESP
- The major operational problems
- Troubleshoot the problems
- Solution suggestions
- What is next

ESP

Problems

- Operational
- Design
- Manufacturer

ESP

Problems

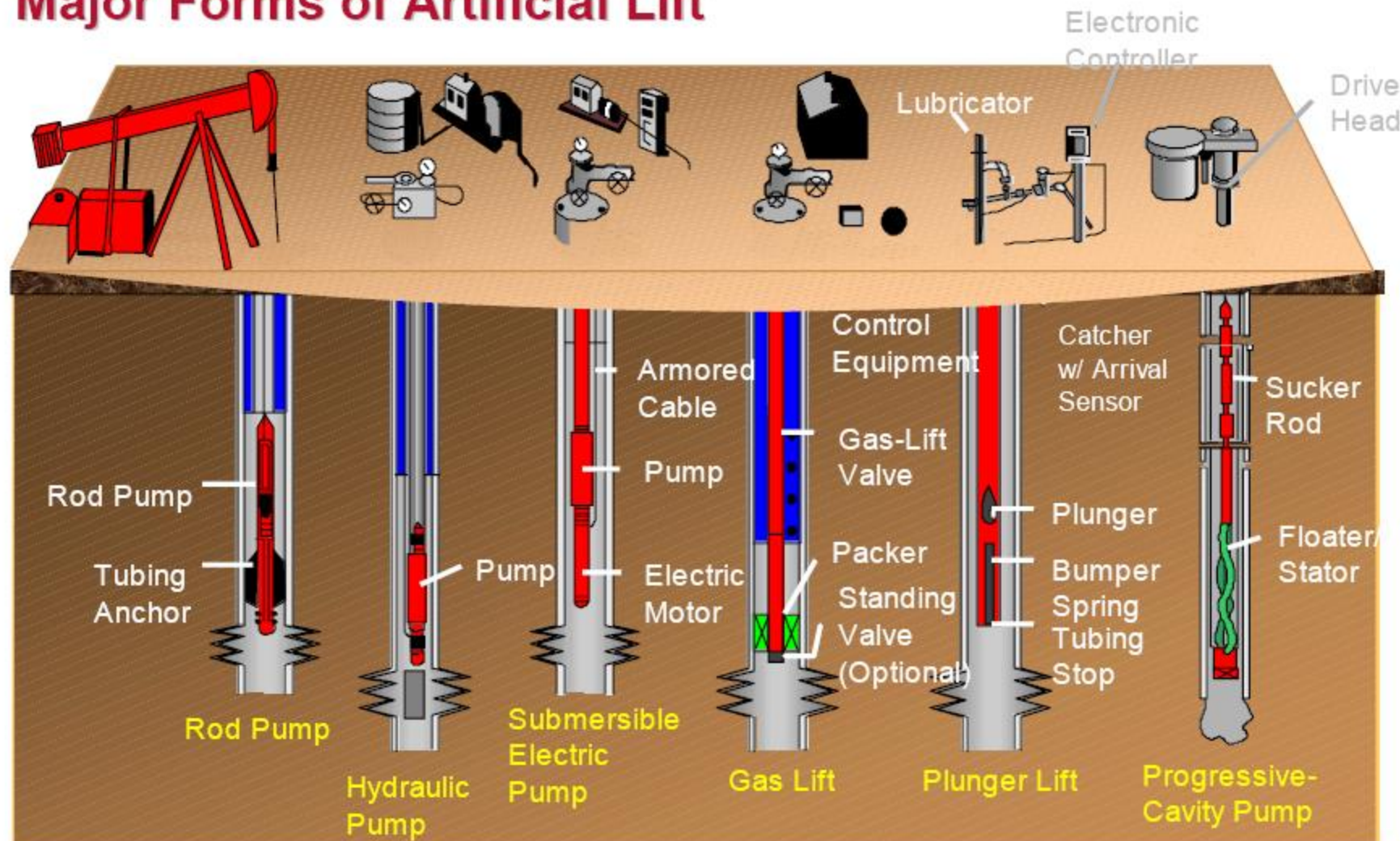
- Symptoms
- Diagnose
- Remedy

ESP PROBLEMS AND MAINTENANCE

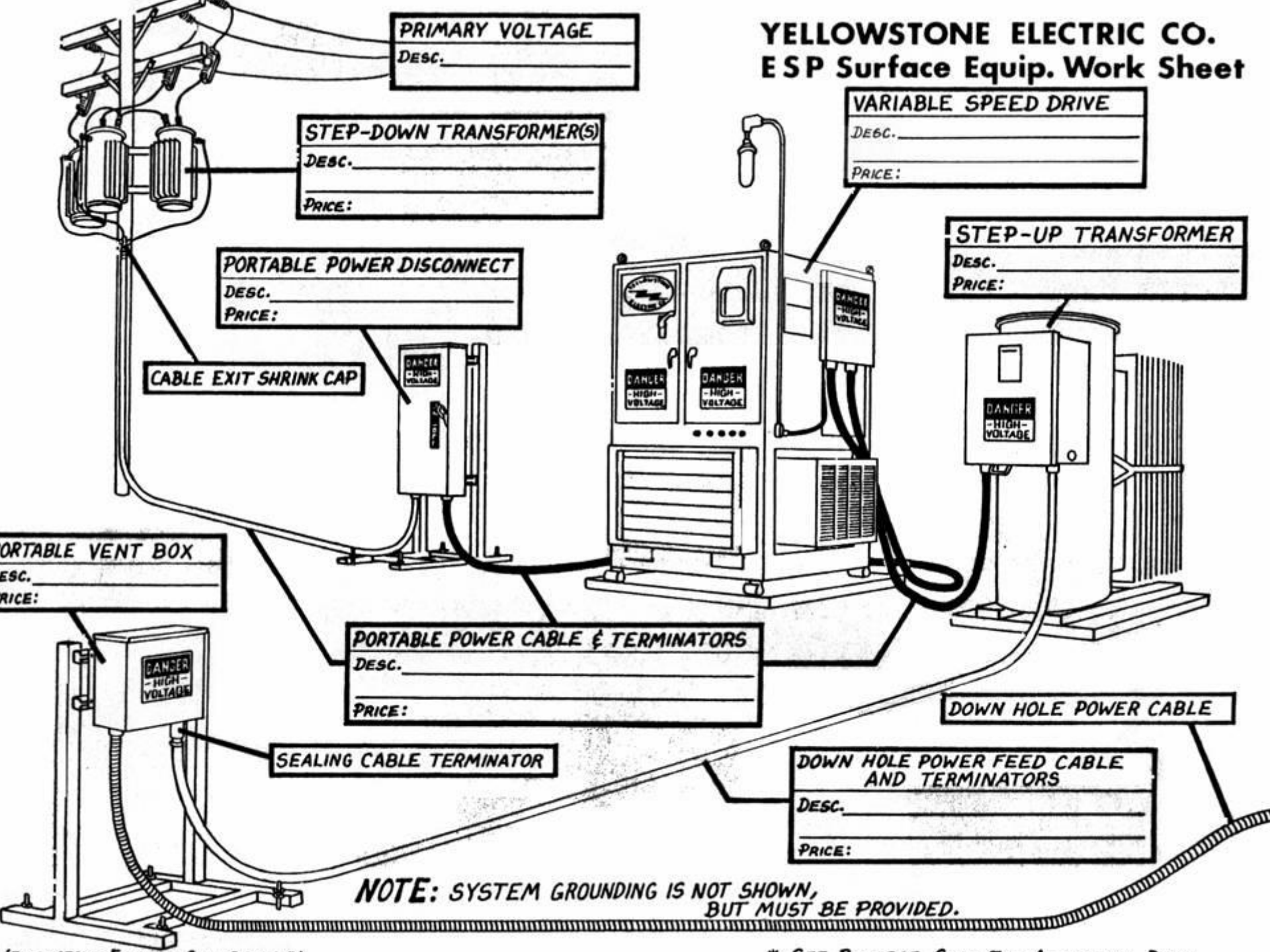
What is Troubleshooting ?

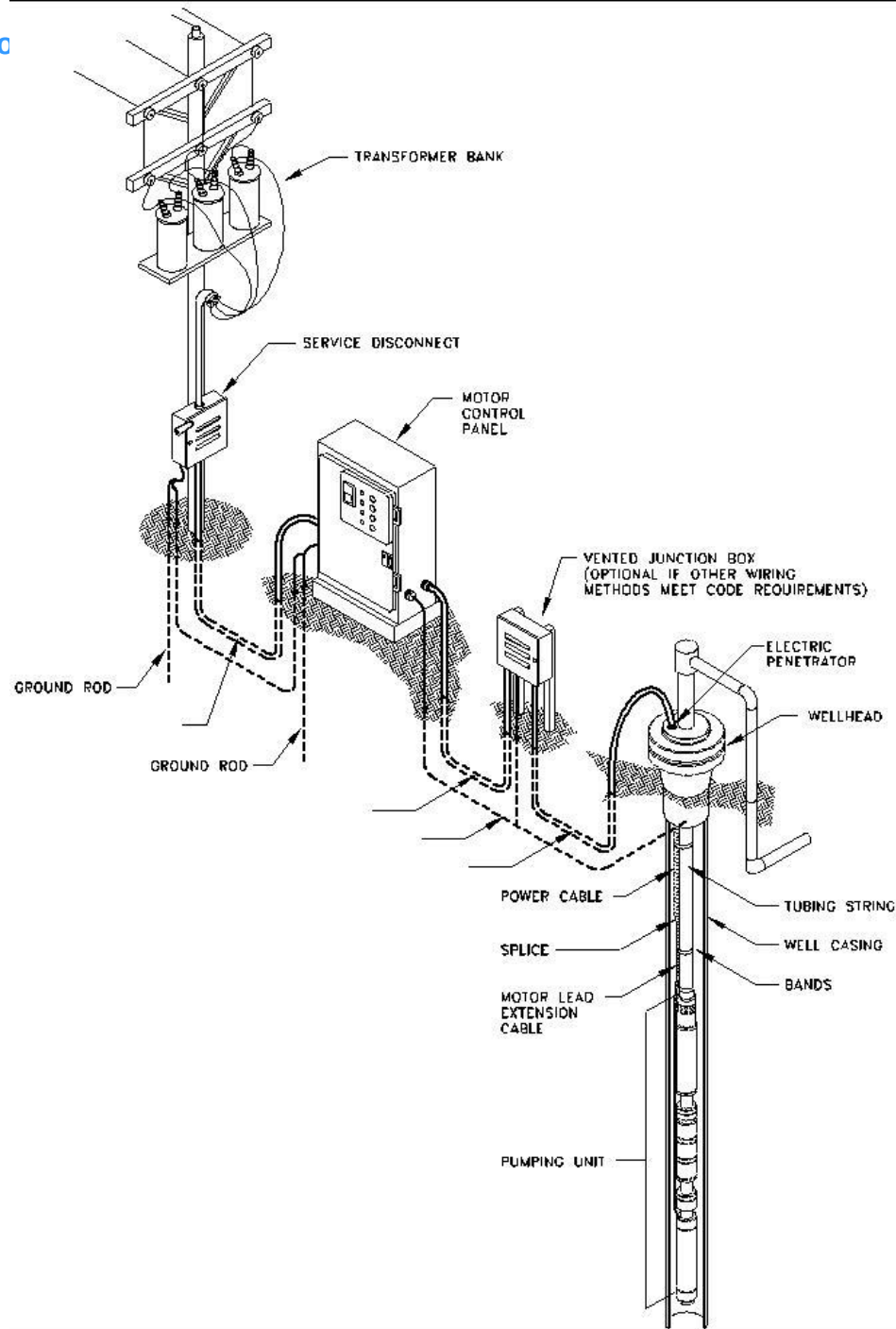
Troubleshooting is the systematic search for the source or causes of a problem so that it can be solved.

Major Forms of Artificial Lift

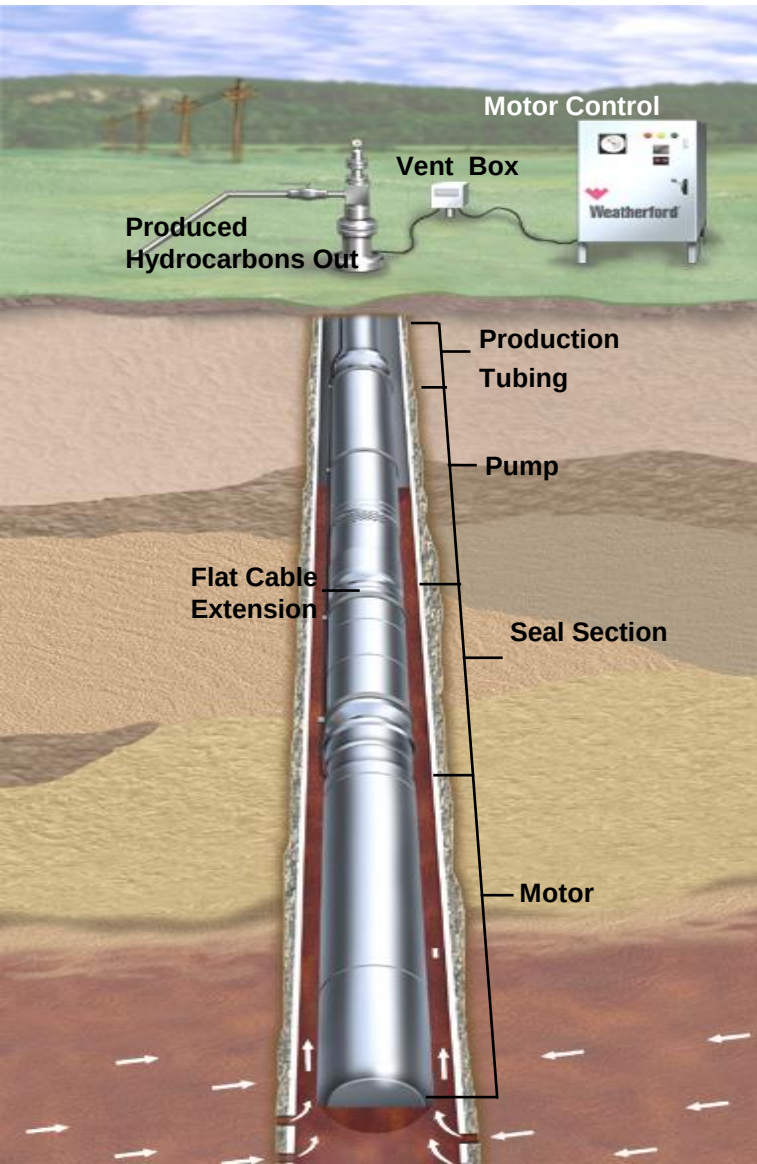


YELLOWSTONE ELECTRIC CO. ESP Surface Equip. Work Sheet





ESP Application Considerations



	Typical Range	Maximum*
Operating Depth	1,000' - 10,000' TVD	15,000' TVD
Operating Volume	200 - 20,000 BPD	30,000 BPD
Operating Temperature	100° - 275° F	400° F
Wellbore Deviation	10°	0 - 90° Pump Placement - <10° Build Angle
Corrosion Handling	Good	
Gas Handling	Poor to Fair	
Solids Handling	Poor to Fair	
Fluid Gravity	>10° API	
Servicing	Workover or Pulling Rig	
Prime Mover Type	Electric Motor	
Offshore Application	Excellent	
System Efficiency	35%-60%	



2. Elimination Process

	Rod Lift	Progressing Cavity	Gas Lift	Plunger Lift	Hydraulic Piston	Hydraulic Jet	Electric Submersible
Operating Depth	100' - 16,000' TVD	2,000' - 6,000' TVD	5,000' - 15,000' TVD	8,000' -	7,500' - 17,000' TVD	5,000' - 15,000' TVD	1,000'- 15,000' TVD
Operating Volume (Typical)	5 - 5000 BPD	5 - 4,500 BPD	200 - 30,000 BPD	1 - 5 BPD	50 - 4,000 BPD	300 - >15,000 BPD	200 - 30,000 BPD
Operating Temperature	100° - 550° F	75°-250° F	100° - 400° F	500° F	100° - 500° F	100° - 500° F	100° - 400° F
Corrosion Handling	Good to Excellent	Fair	Good to Excellent	Excellent	Good	Excellent	Good
Gas Handling	Fair to Good	Good	Excellent	Excellent	Fair	Good	Poor to Fair
Solids Handling	Fair to Good	Excellent	Good	Fair	Poor	Good	Poor to Fair
Fluid Gravity	>8° API	<35° API	>15° API	GLR Required 300 SCF/BBL/	>8° API	>8° API	>10° API
Servicing	Workover or Pulling Rig	Workover or Pulling Rig	Wireline or Workover Rig	Wellhead Catcher or Wireline	Hydraulic or Wireline	Hydraulic or Wireline	Workover or Pulling Rig
Prime Mover	Gas or Electric	Gas or Electric	Compressor	Wells' Natural Energy	Multicylinder or Electric	Multicylinder or Electric	Electric Motor
Offshore Application	Limited	Good	Excellent	N/A	Good	Excellent	Excellent
Overall System Efficiency	45% - 60%	40% - 70%	10% - 30%	N/A	45% - 55%	10% - 30%	35% - 60%

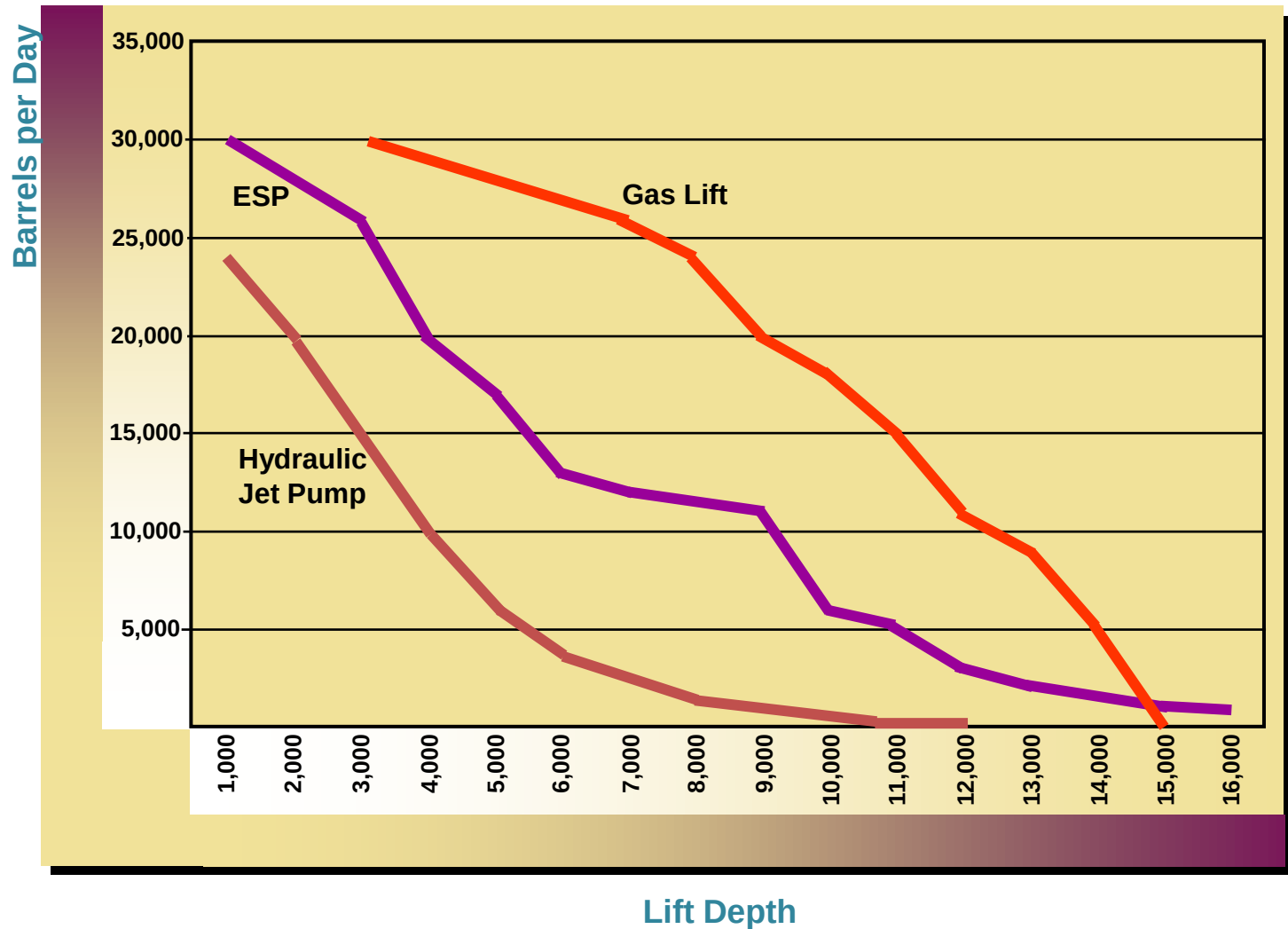


2. Elimination Process

	Rod Lift	Progressing Cavity	Gas Lift	Plunger Lift	Hydraulic Piston	Hydraulic Jet	Electric Submersible
Operating Depth	100' - 16,000' TVD	2,000' - 6,000' TVD	5,000' - 15,000' TVD	8,000' - 19,000' TVD	7,500' - 17,000' TVD	5,000' - 15,000' TVD	1,000' - 15,000' TVD
Operating Volume (Typical)	5 - 5000 BPD	5 - 4,500 BPD	200 - 30,000 BPD	1 - 5 BPD	50 - 4,000 BPD	300 - >15,000 BPD	200 - 30,000 BPD
Operating Temperature	100° - 550° F	75°-250° F	100° - 400° F	120° - 500° F	100° - 500° F	100° - 500° F	100° - 400° F
Corrosion Handling	Good to Excellent	Fair	Good to Excellent	Excellent	Good	Excellent	Good
Gas Handling	Fair to Good	Good	Excellent	Excellent	Fair	Good	Poor to Fair
Solids Handling	Fair to Good	Excellent	Good	Poor to Fair	Poor	Good	Poor to Fair
Fluid Gravity	>8° API	<35° API	>15° API	GLR Required - 300 SCF/BBL/ 1000' Depth	>8° API	>8° API	>10° API
Servicing	Workover or Pulling Rig	Workover or Pulling Rig	Wireline or Workover Rig	Wellhead Catcher or Wireline	Hydraulic or Wireline	Hydraulic or Wireline	Workover or Pulling Rig
Prime Mover	Gas or Electric	Gas or Electric	Compressor	Wells' Natural Energy	Multicylinder or Electric	Multicylinder or Electric	Electric Motor
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2. Elimination Process

High
Volume
Hydraulic
Jet Pumps,
ESP
Gas lift

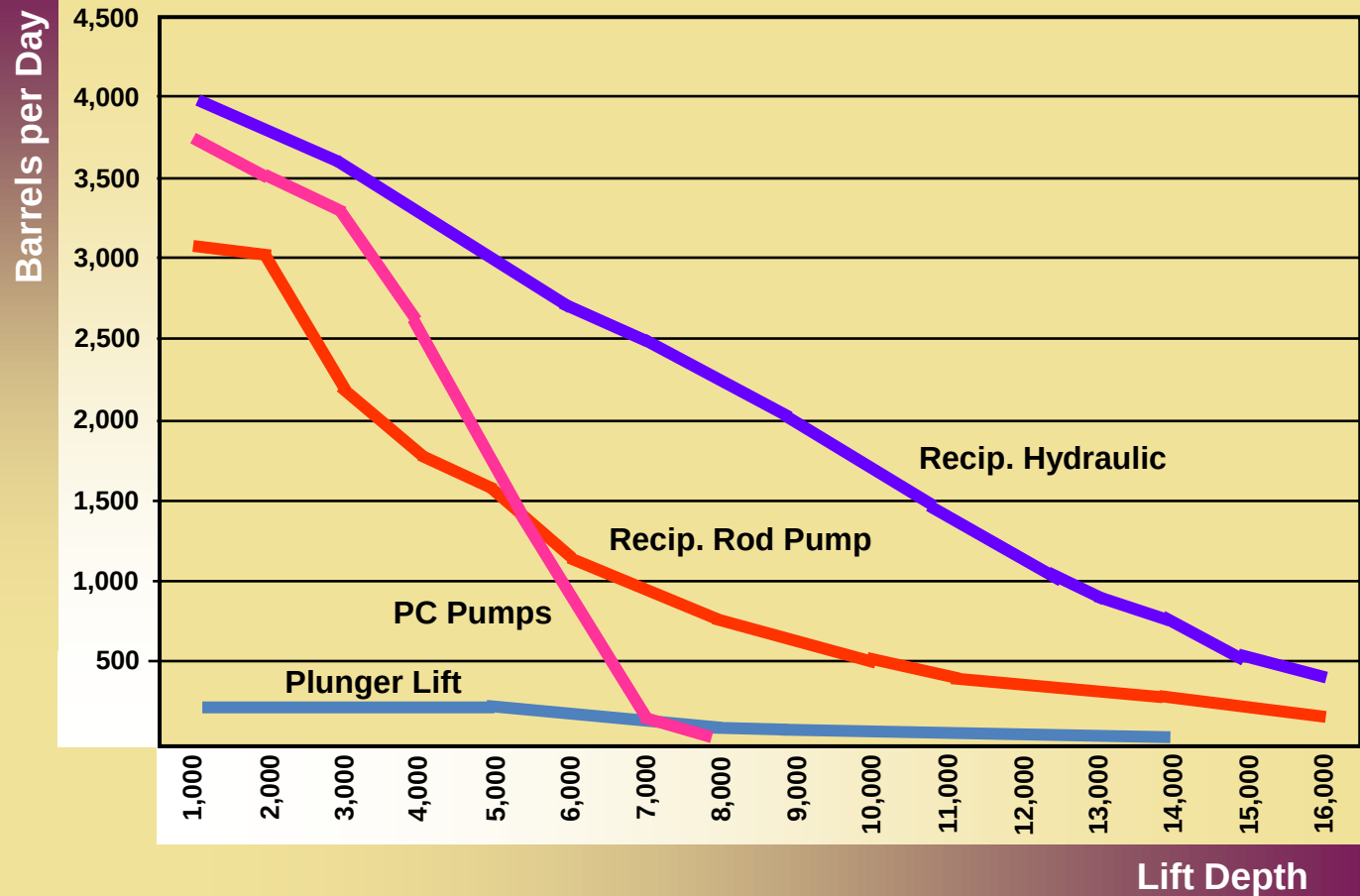




2. Elimination Process

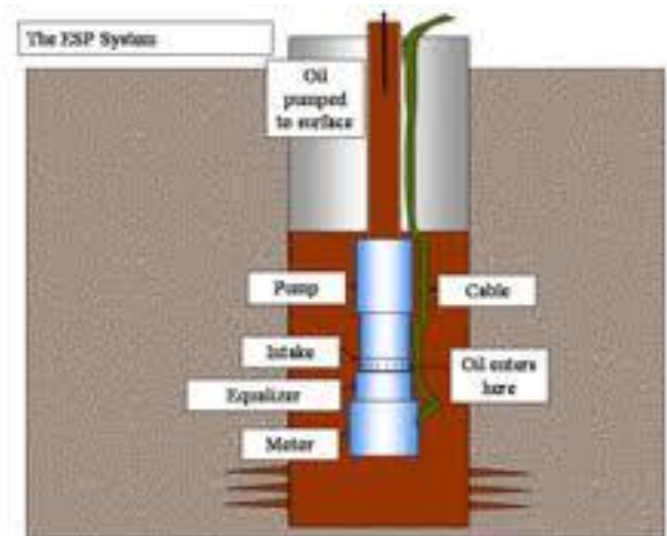
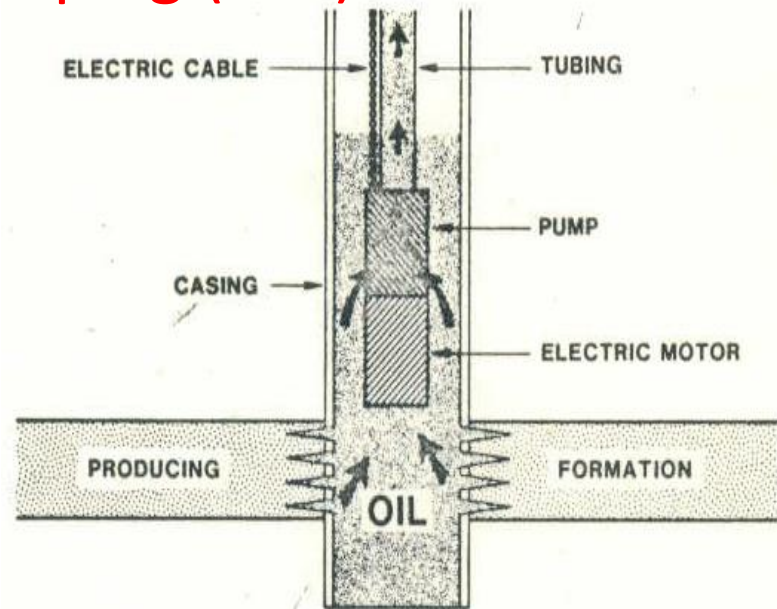
**Lower
Volume**

**Reciprocating
Hydraulic
Pumps,
PC Pumps,
Rod Pumps &
Plunger Lift**

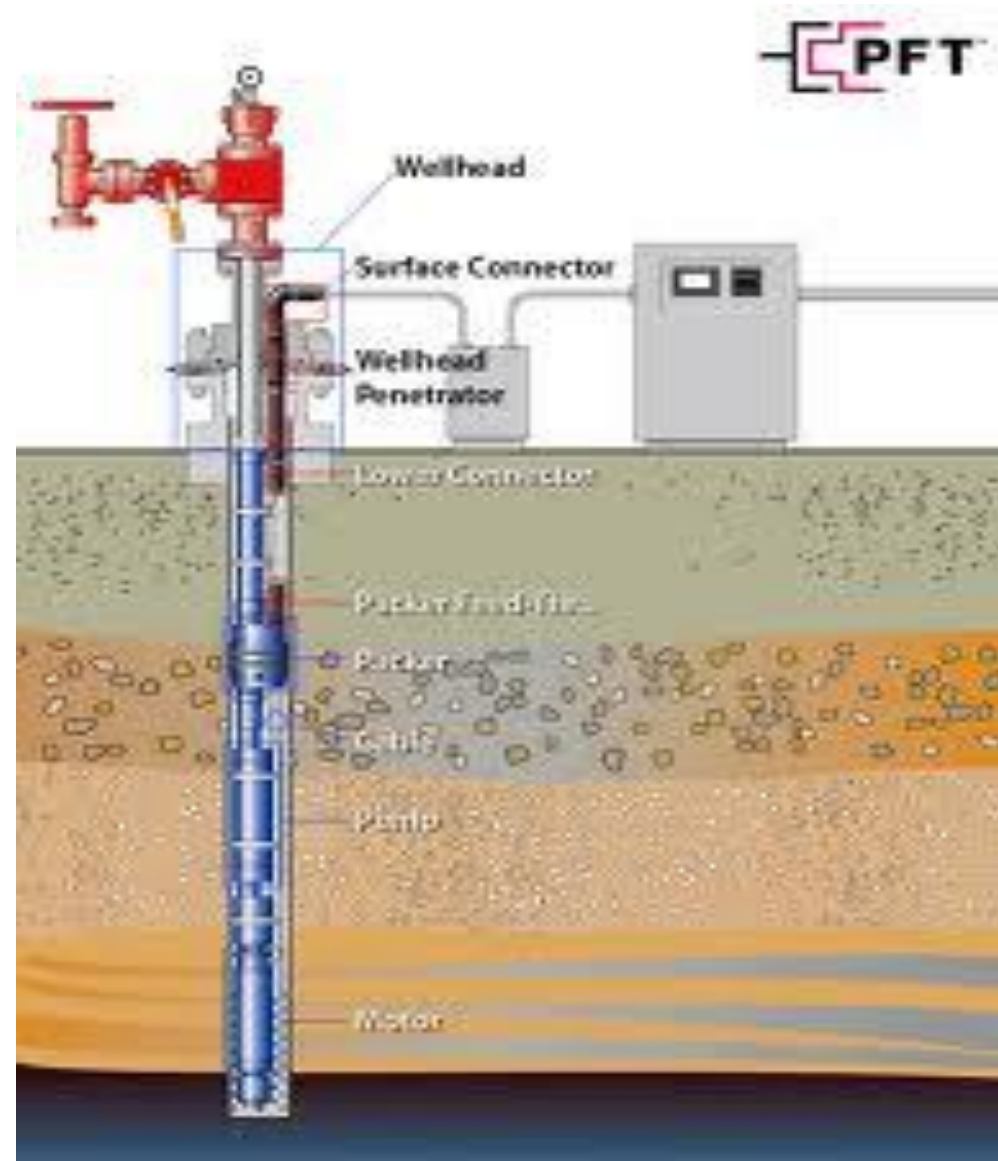


Artificial Lift - Submersible Pumping (ESP)

- Consists of an electric motor attached to a pump on the end of the tubing string.
- The electric motor turns a centrifugal pump which forces oil from the bottom of the well, up through the inside of the tubing, and out at the surface.
- The electricity is supplied through an electric cable attached to the side of the tubing and connected to the electric motor.



Artificial Lift - Submersible Pumping (ESP)



ESP

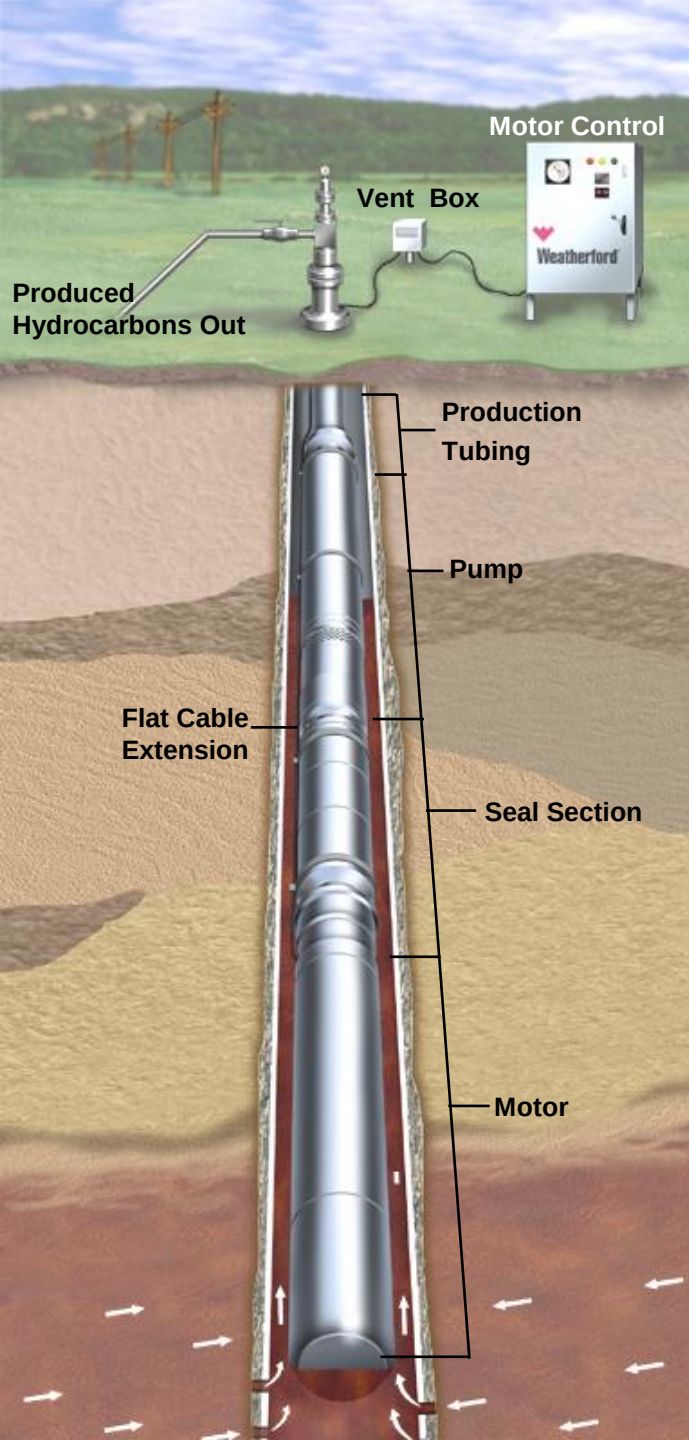
Advantages:

1. The most efficient lift methods on a cost-per-barrel basis.
2. High rate: 100 to 60,000 B/D, including high water-cut fluids.
3. Work in high-temperature wells (<400 F) using high-temperature motors and cables.
4. The pumps can be modified to lift corrosive fluids and sand.
5. ESP systems can be used in high-angle and horizontal wells if placed in straight or vertical sections of the well.

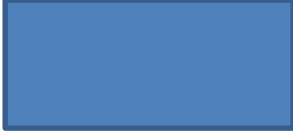


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ESP more Advantages



- 💧 **Low Maintenance**
- 💧 **Minor Surface Equipment Needs**
- 💧 **Adaptable to All Wells With 4-1/2" Casing and Larger**
- 💧 **Use for Well Testing**
- 💧 **Minimal surface equipment requirements.**



More Benefits of ESPs

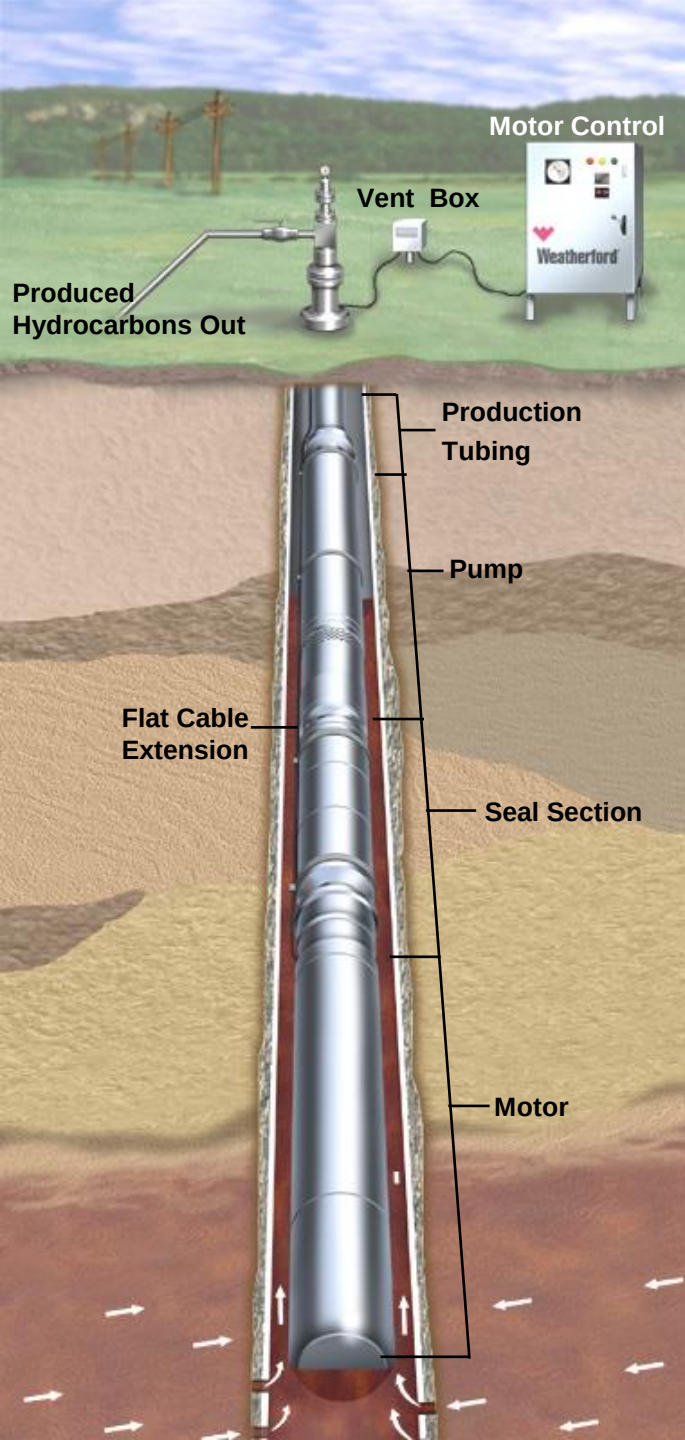
- a) They can be economically designed for both oil and water wells, up to 15,000 feet.
- b) They have a relatively small “surface footprint,” and so are appropriate for use in offshore, urban or other confined locations. They are relatively simple to operate.
- c) They generally provide low lifting costs for high fluid volumes.
- d) They make it easy to apply corrosion and scale treatments.



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ESP Limitations

- 💧 Available Electric Power
- 💧 Limited Adaptability to Major Changes in Reservoir
- 💧 Difficult to Repair In the Field
- 💧 Free Gas and/or Abrasives
- 💧 High Viscosity
- 💧 Higher Pulling Costs
 - **Poor ability to pump sand**



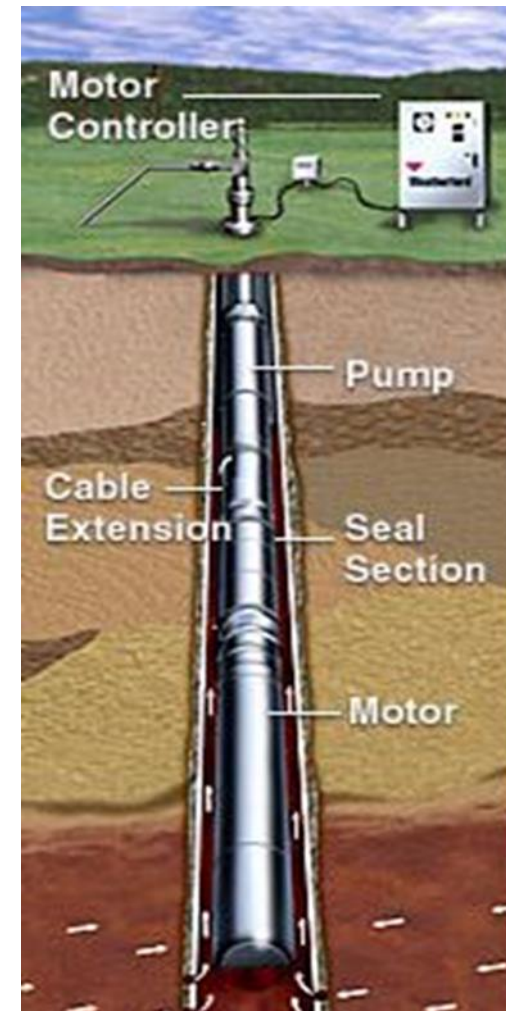
ESP

Disadvantages:

ESP pumps can be damaged from “gas lock”.

In wells producing high GOR fluids, a downhole gas separator must be installed.

ESP pumps have limited production ranges determined by the number and type of pump stages; changing production rates requires either a pump change or installation of a **variable-speed surface drive**. The tubing must be pulled for pump repairs or replacement.



ESP



Problems with sand production, high gas liquid ratio and high bottom hole temperatures

Limitations of ESPs

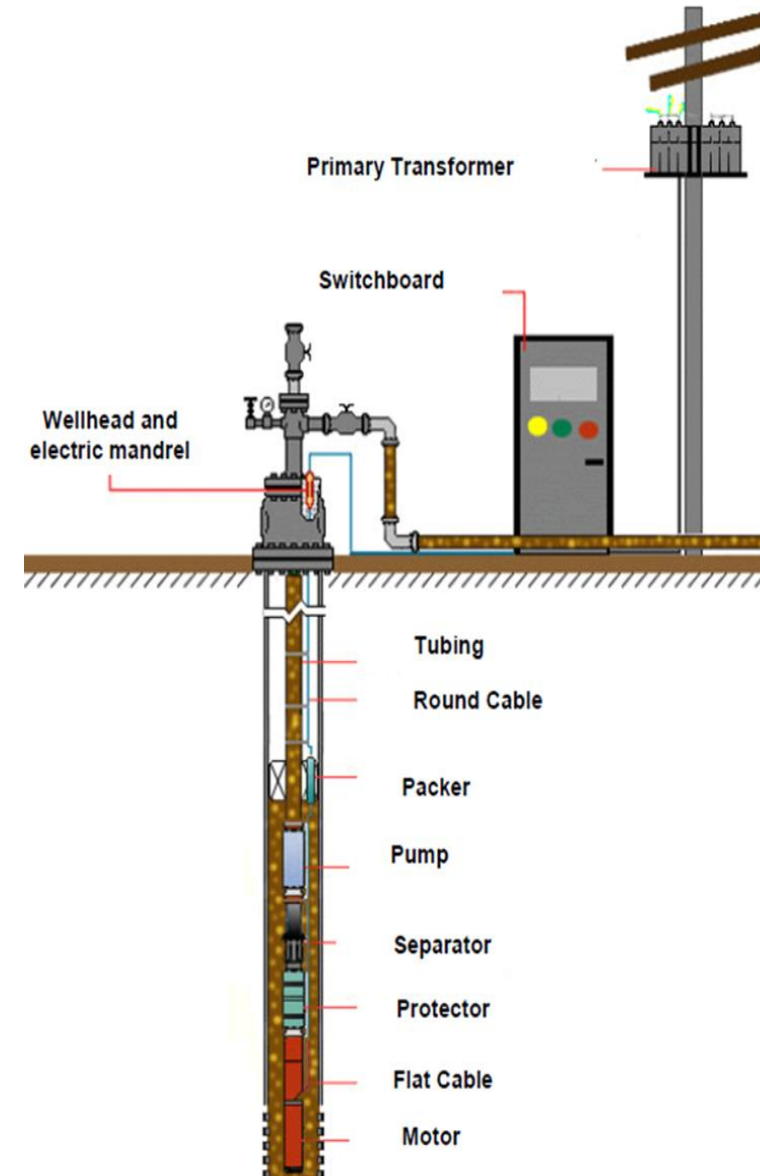
- a) They are generally limited to single-zone completions
- b) They requires a source of high-voltage electric power
- c) The presence of a power cable alongside the tubing string can make it more difficult to run or pull tubing.
- d) They are not particularly good at handling gas and solids production.
- e) Analyzing the system performance can be a challenge.
- f) Power cables may deteriorate in high temperature conditions—400 F (about 200 C) is their general upper limit with respect to operating temperature.

ESP

The ESP **downhole assembly** is located in the well at the bottom of the tubing.

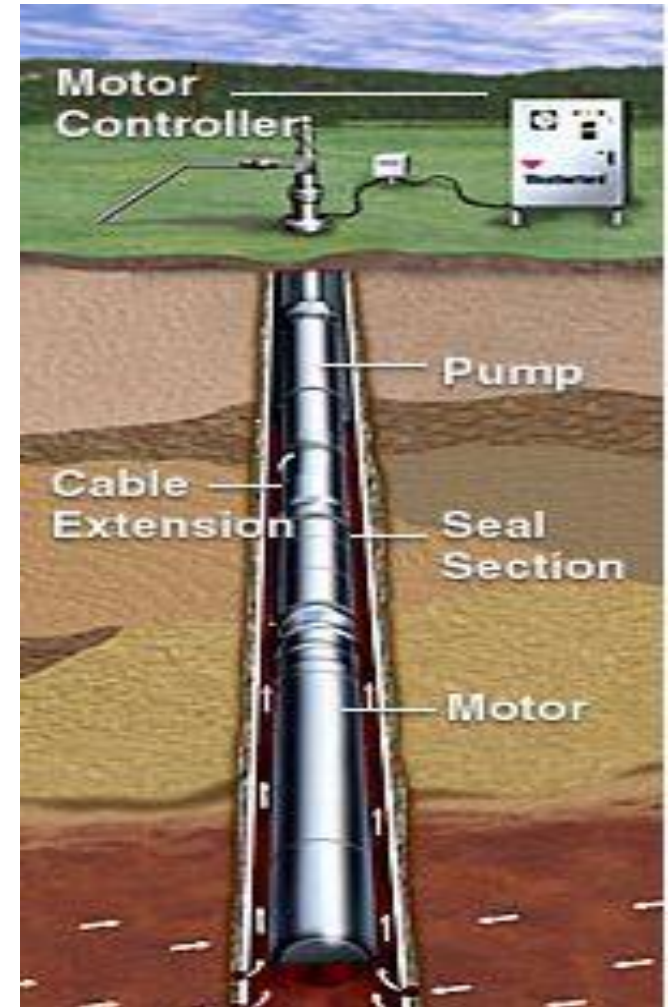
The motor, seal, protector ; intake and ; pump assembly,

- intake that allows fluid to enter the pump.
- a *protector* or *seal*, which equalizes internal and external pressures and protects the motor from well fluids.
- motor that drives the pump.
- The assembly is positioned in the well above the perforations; this allows fluid entering the intake to flow past the motor and cool it.



ESP

(ESP) assembly consists of a multi-stage centrifugal pump driven by a submersible electric motor, which is connected to a power source at the surface

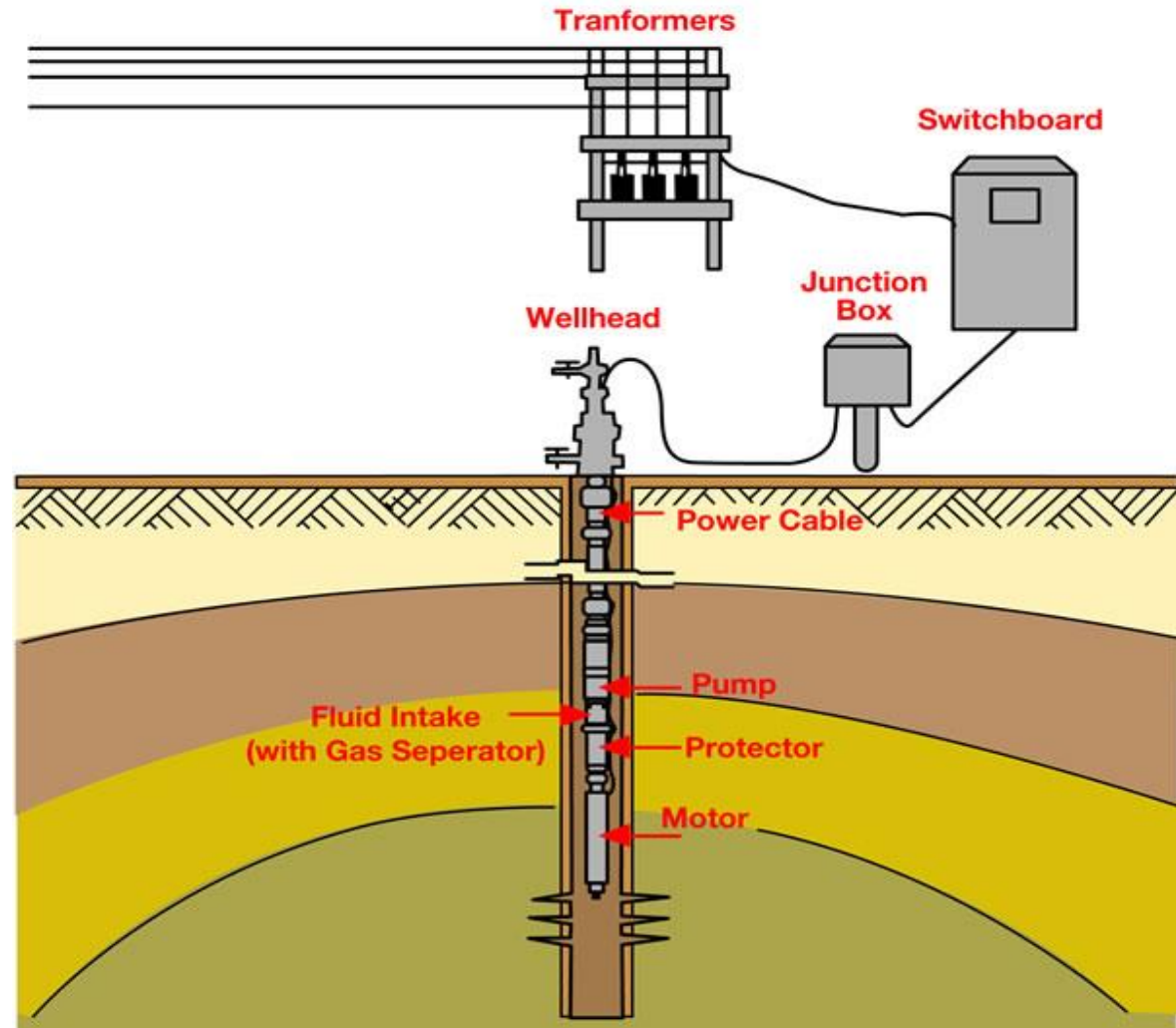


ESP

surface equipment includes:

- transformers,
- switchboard,
- junction box
- surface power cables.

Power passes through a cable running from the transformer to the switchboard and junction box, then to the wellhead



Introduction to ESP

ESP Components

Power passes through a cable running from the transformer to the switchboard and junction box, then to the wellhead.

The motor, seal, intake and pump assembly, along with the power cable, goes in the well as the tubing is run. The well power cable is spliced to a motor cable that is connected to the outside of the downhole assembly.

Below the pump is an intake that allows fluid to enter the pump.

Below the intake is a gas separator and a *protector* or *seal*, which equalizes internal and external pressures and protects the motor from well fluids.

At the bottom is a motor that drives the pump. The assembly is positioned in the well above the perforations; this allows fluid entering the intake to flow past the motor and cool it.

Introduction to ESP

ESP Components

An ESP system can be divided into two categories:

- **Surface components**

- Transformers (Primary and Secondary)
- Switchboard or Variable Speed Drive or Soft Start
- Junction Box
- Wellhead

- **Subsurface components**

- Cable
- Cable Guards
- Cable Clamps
- Pump
- Gas Separator (Optional)
- Seal Section
- Motor
- Sensor (Optional)
- Drain Valve
- Check Valve

Introduction to ESP

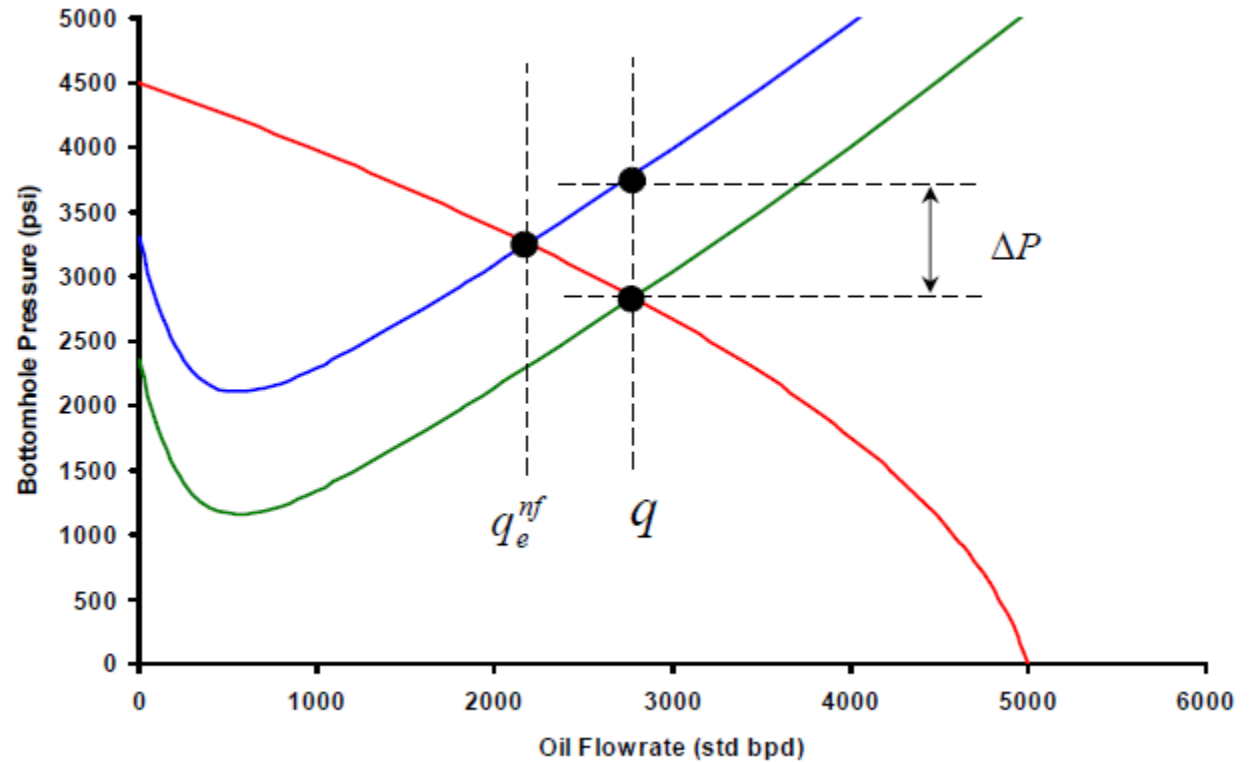
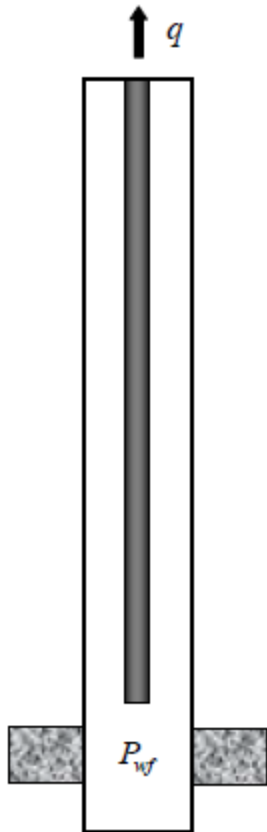
Principles of an ESP

For a naturally flowing well the intersection of the IPR and OPR curves defines the natural equilibrium flowrate

- For a naturally flowing well it is possible to produce a wide range of flow rates smaller than the naturally flowing flowrate with the use of a choke
- On the other hand, in a naturally flowing well without artificial lift equipment, production flowrates higher than the natural flowrate are impossible to be achieved since for those conditions, the OPR pressures are bigger than the IPR pressures

Introduction to ESP

Principles of an ESP



Introduction to ESP

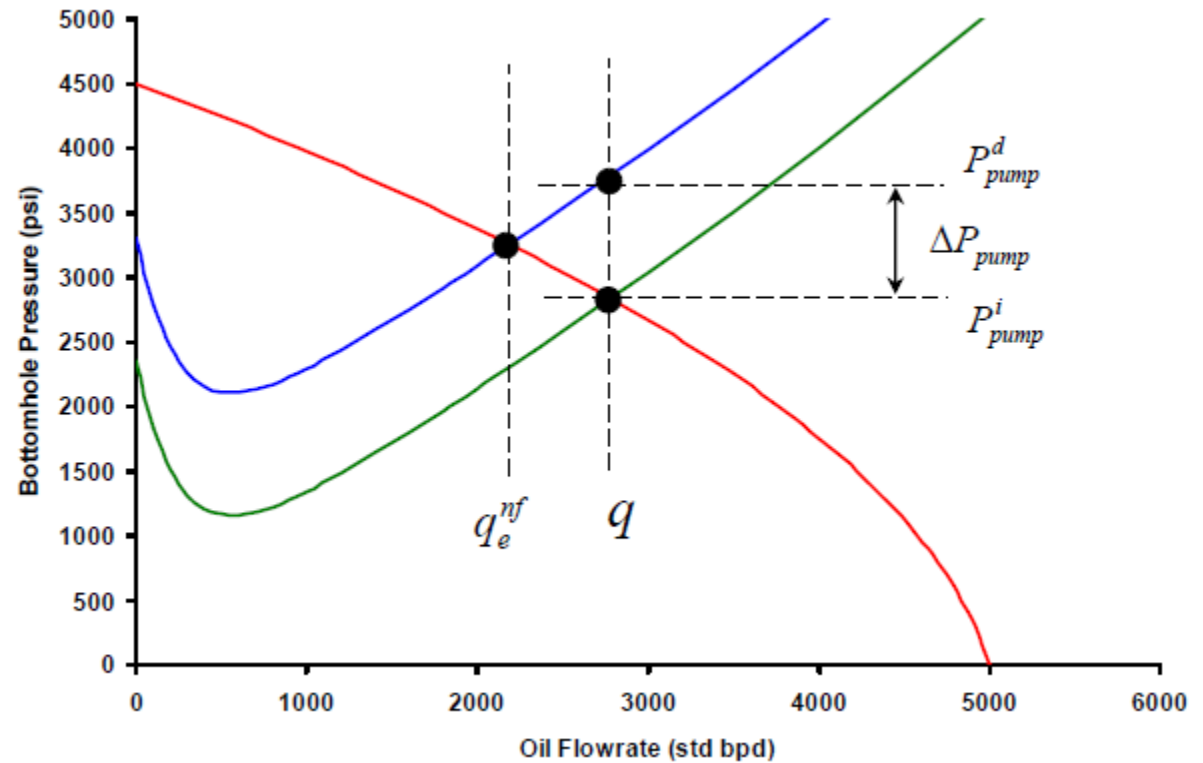
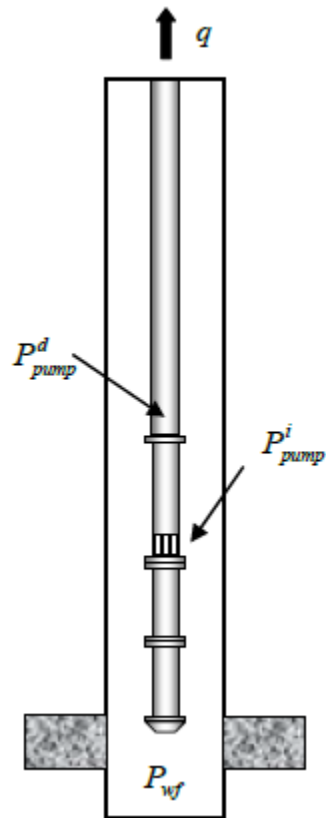
Principles of an ESP

In order to produce flowrates higher than the natural equilibrium flowrate the use of an artificial lift system is necessary

- If an ESP is installed in the tubing string close to the perforations, the discharge pressure of the pump must be equal to the OPR pressure and the intake pressure of the pump must be equal to the IPR pressure**
- The difference between the OPR and IPR bottom hole flowing pressure for flowrates bigger than the natural equilibrium flowrate defines the pressure increment that the ESP must deliver**

Introduction to ESP

Principles of an ESP



Introduction to ESP

Principles of an ESP

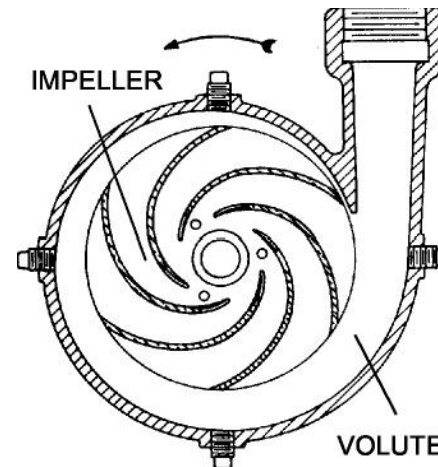
The submersible pumps are multistage centrifugal pumps operating in a vertical position.

Produced liquids, after being subjected to great centrifugal forces caused by the high rotational speed of the impeller, lose their kinetic energy in the diffuser where a conversion of kinetic to pressure energy takes place. This is the main operational mechanism of radial and mixed flow pumps.

The ratio between the centrifugal force and the gravitational force:

$$\frac{F_c}{F_g} = \frac{mu^2/r}{mg} = \frac{u^2}{rg} = \frac{\omega^2 r}{g}$$

If $w = 3600$ RPM, $r = 4''$ then this ratio is 131,673



Introduction to ESP

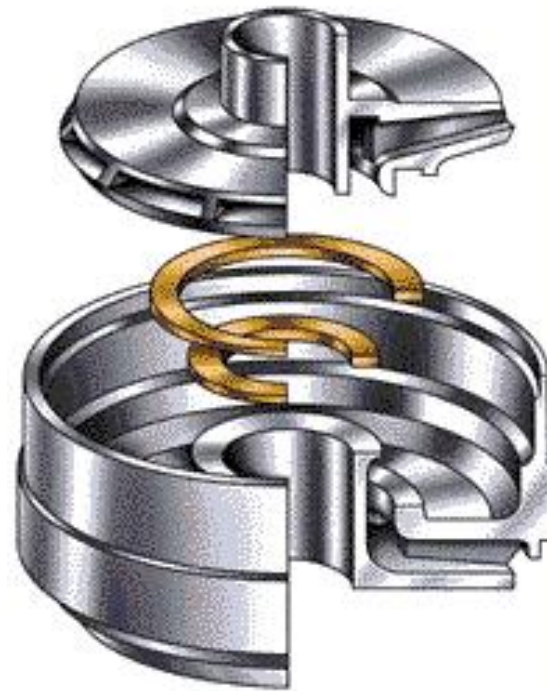
ESP Classifications

ESP centrifugal stages are classified according to their design as:

- Radial stages
- Mixed flow stages



Mixed flow stages



Radial stages

Introduction to ESP

ESP Classifications

The performance characteristics of stages at the best efficiency point is a function of a dimensionless number called **specific speed**

$$N_s = \frac{N \sqrt{q_{bep}}}{H_{bep}^{3/4}}$$

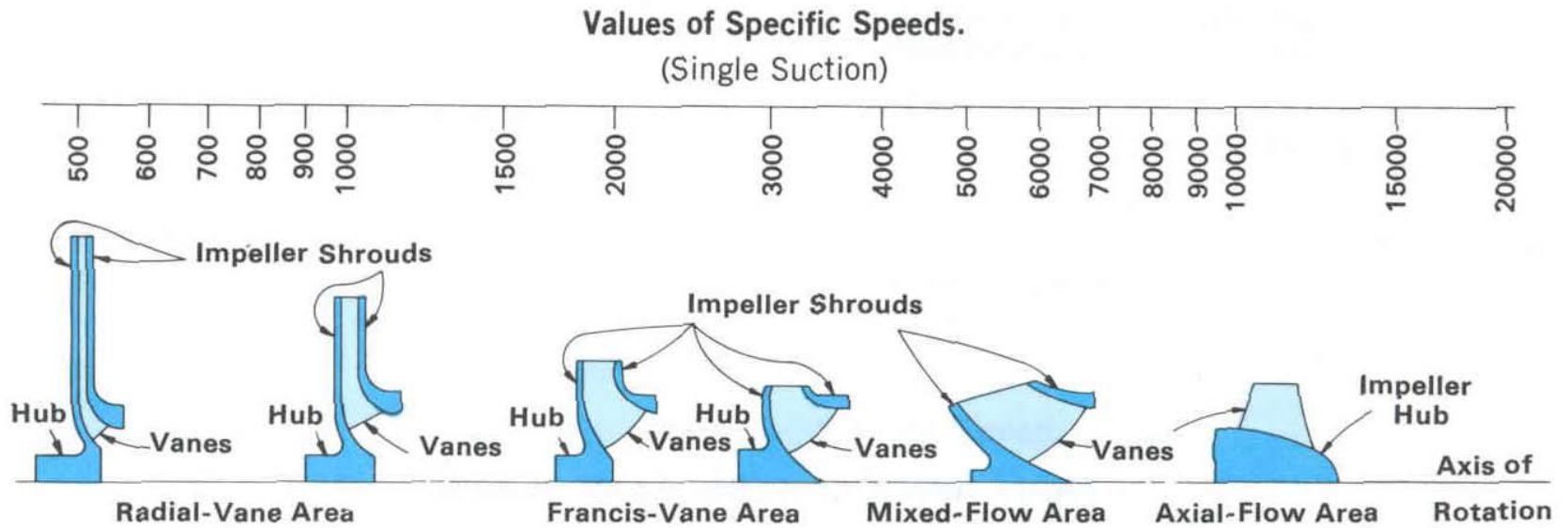
rpm → N gpm → q_{bep}
ft → H_{bep}

$$N_s = 0.1707825 \frac{N \sqrt{q_{bep}}}{H_{bep}^{3/4}}$$

rpm → N bpd → q_{bep}
ft → H_{bep}

Introduction to ESP

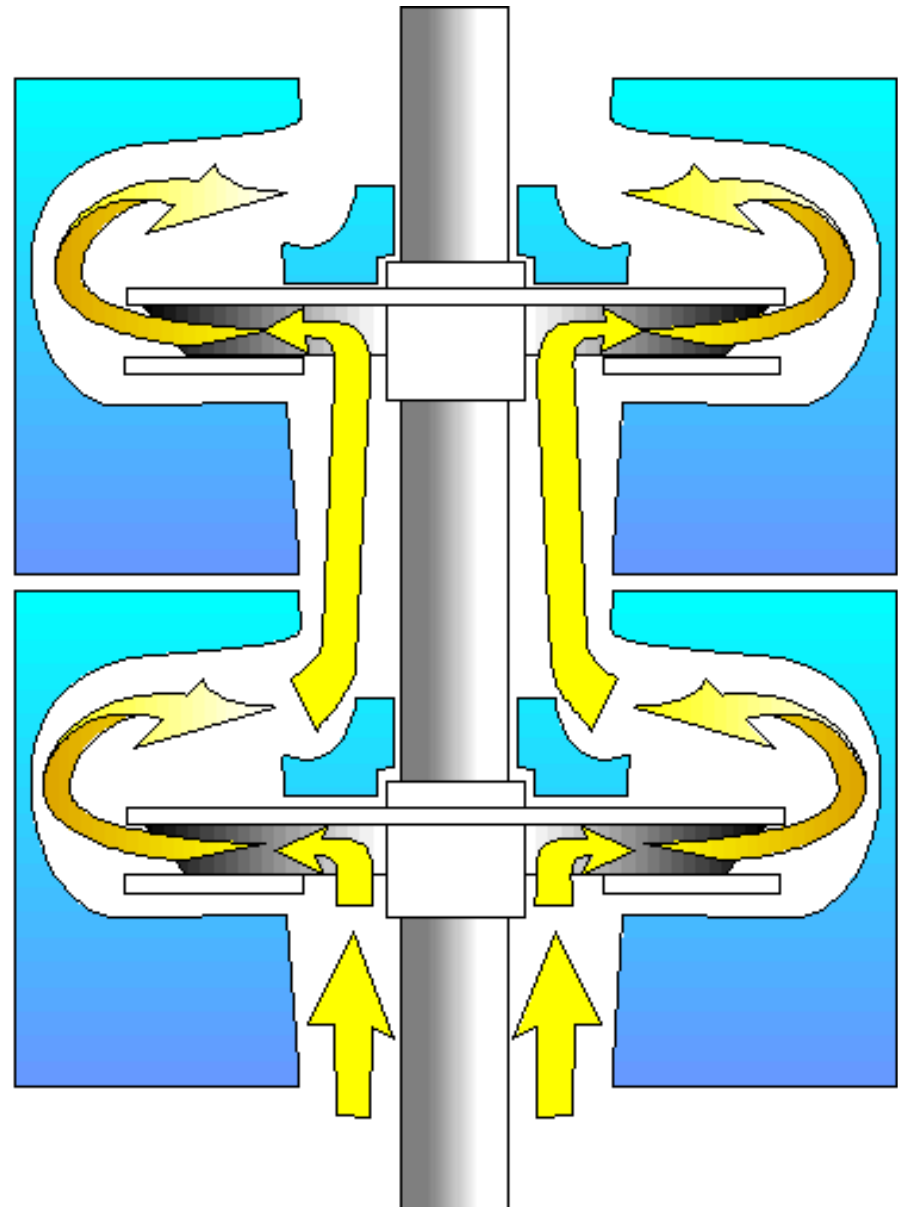
ESP Classifications



$$N_s = \frac{N \sqrt{q_{bep}}}{H_{bep}^{3/4}}$$

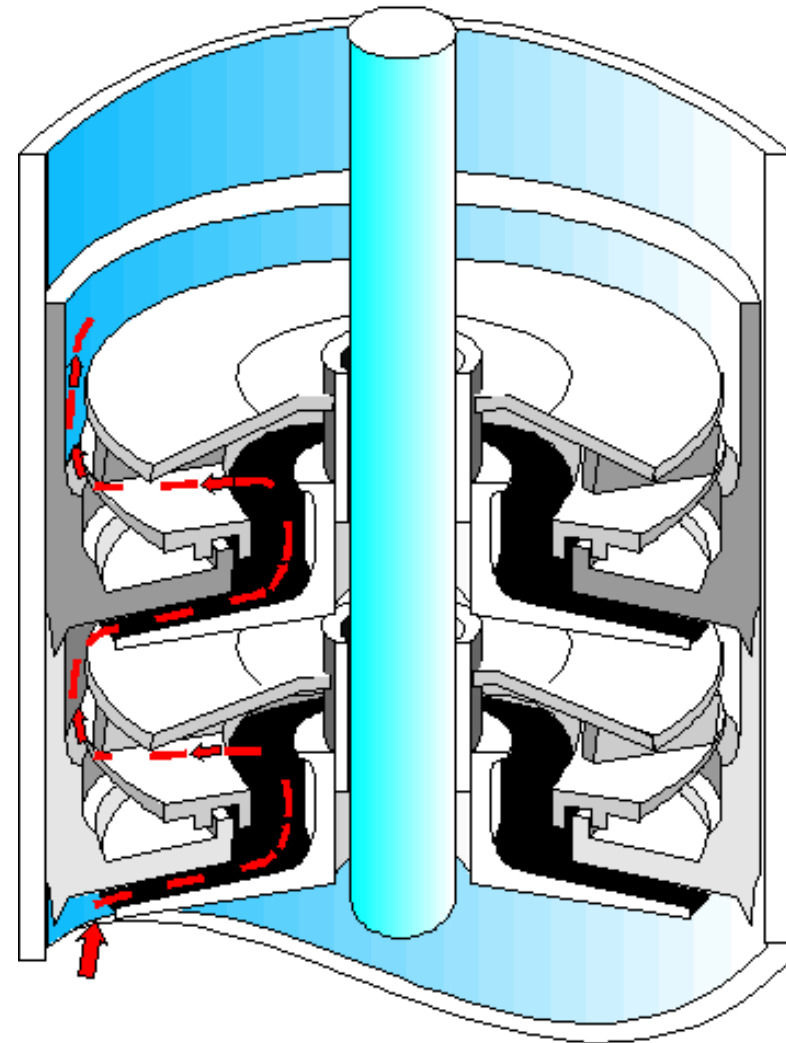
Pump Stage

- Fluid enters impeller through 'eye' near shaft and exits impeller on OD
- Diffuser (in blue) redirects fluid into next impeller



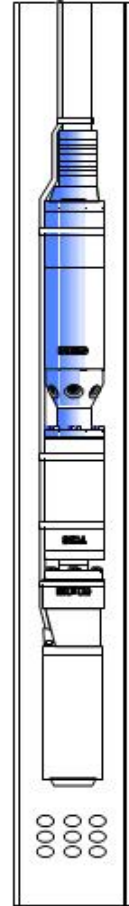
Pump Stage

- Cut-away view of two impellers and diffusers
- By stacking impellers and diffusers (multi-staging), desired lift (TDH) is achieved



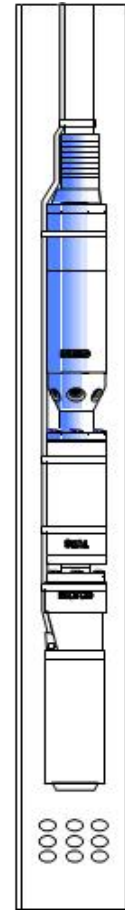
Downhole Centrifugal Pump

- Centrifugal pump is so named because head added to fluid is due largely to centrifugal effects
- Characterized by:
 - Small diameter
 - Large number of stages
 - High design loads

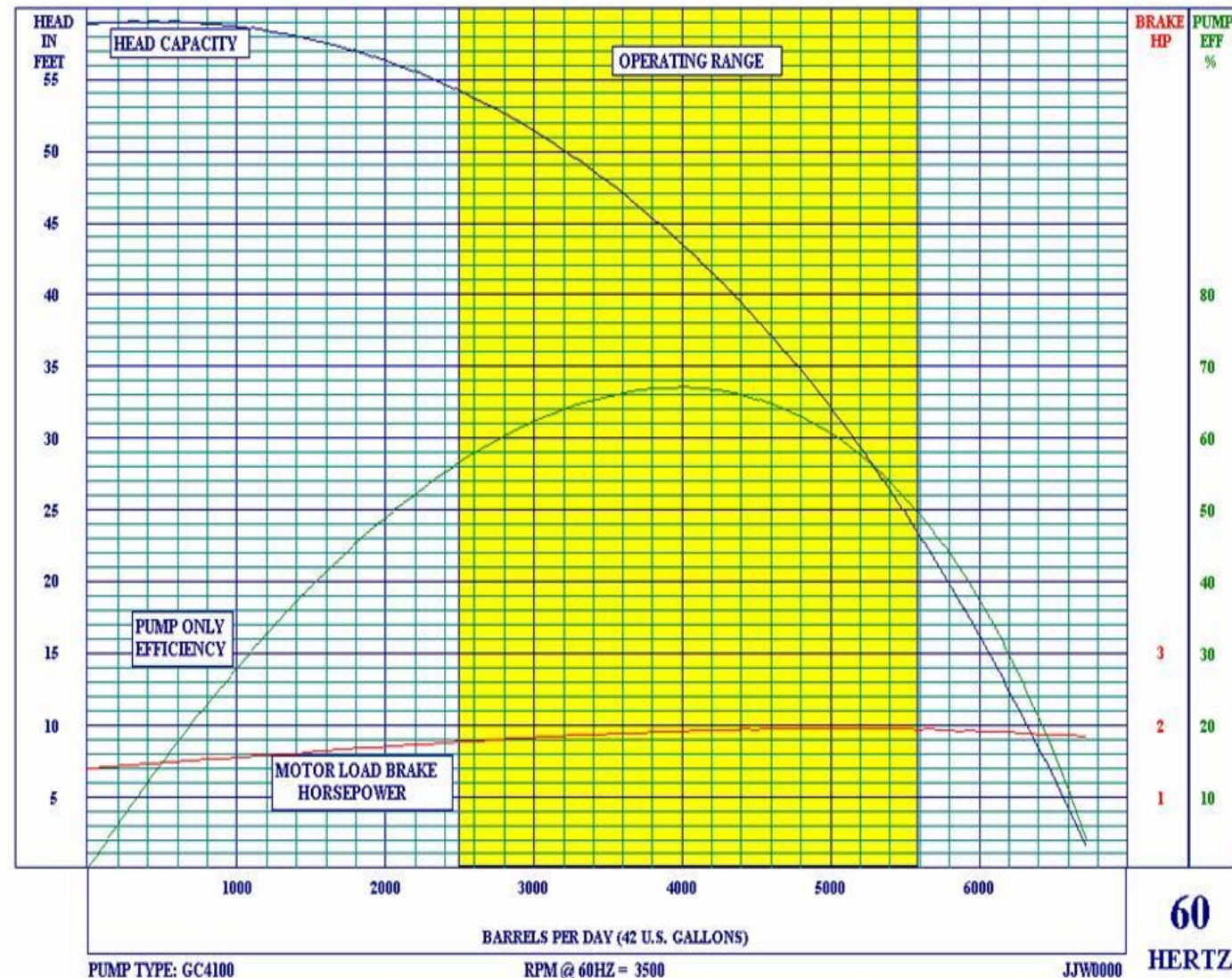


Downhole Centrifugal Pump

- Normally driven by a two-pole submersible motor at an operating speed of 3,500 rpm
- Performance:
 - Head = $f(\mu, \text{glr})$
 - Flow = $f(\mu)$
 - Power = $f(s_g, \mu)$



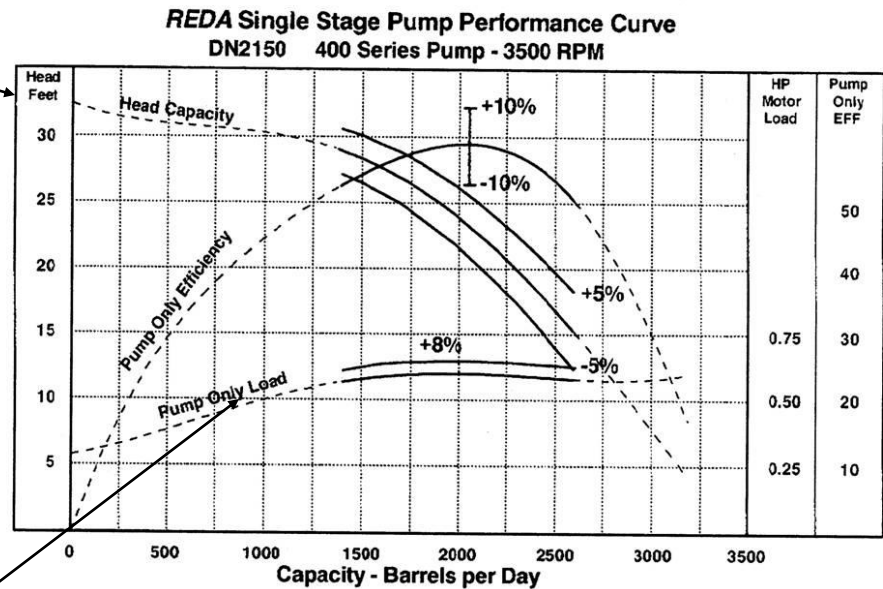
A5000 ft oil well
producing with ESP
3000bpd
Determine
1. discharge head ft
2. Efficiency
3. HP
4. No of stages



GC4100

Typical Performance Curve

$$Head = \frac{\Delta p}{\rho} = \frac{\Delta p}{62.4 \times \gamma / 144} = \frac{\Delta p}{.433 \times \gamma}$$

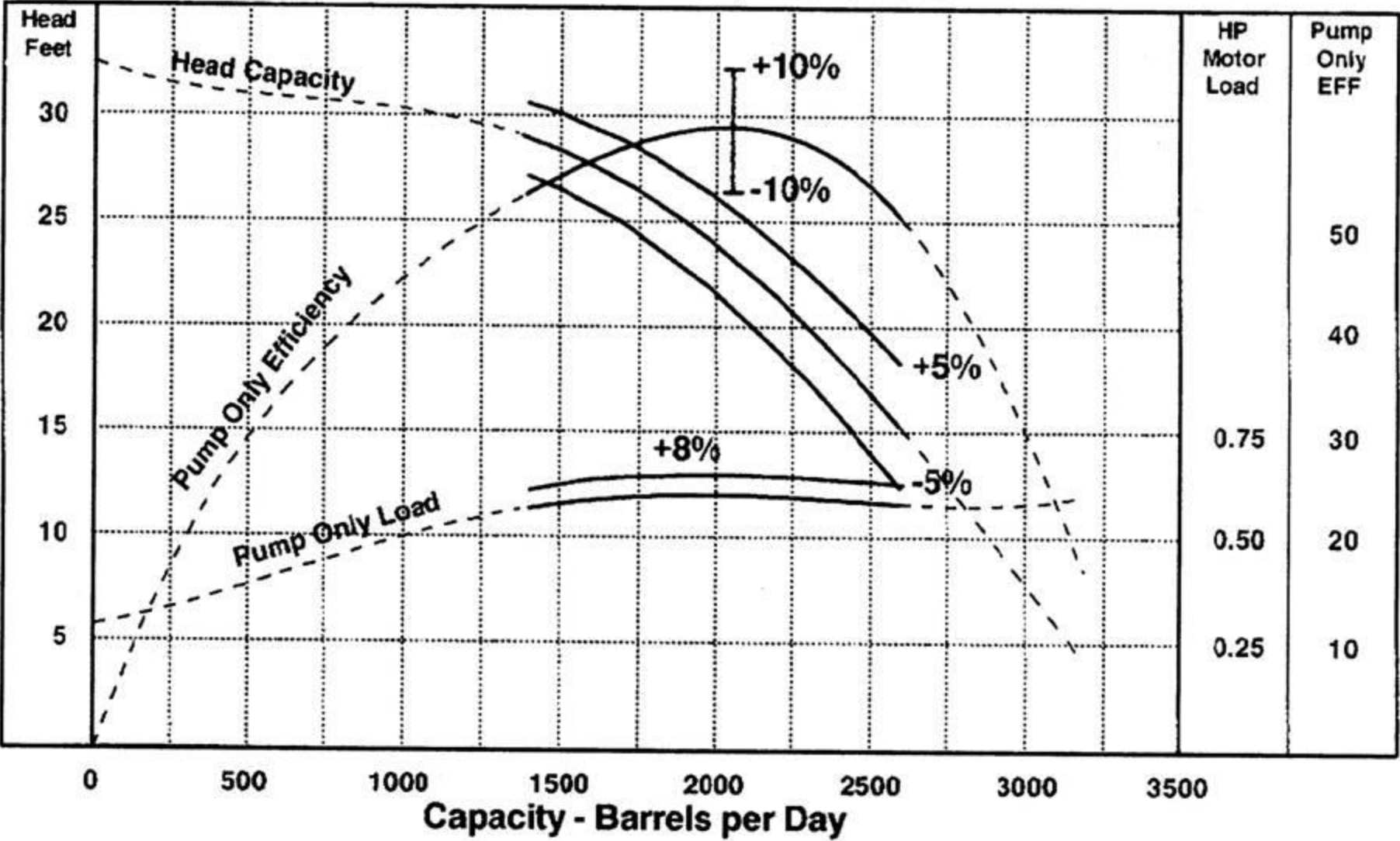


Simple Sizing
Example will be
done here

$$HP = HP \text{ from Chart} \times Sp.Gr.$$

TEST CURVE SHOWING API TOLERANCES

REDA Single Stage Pump Performance Curve
DN2150 400 Series Pump - 3500 RPM



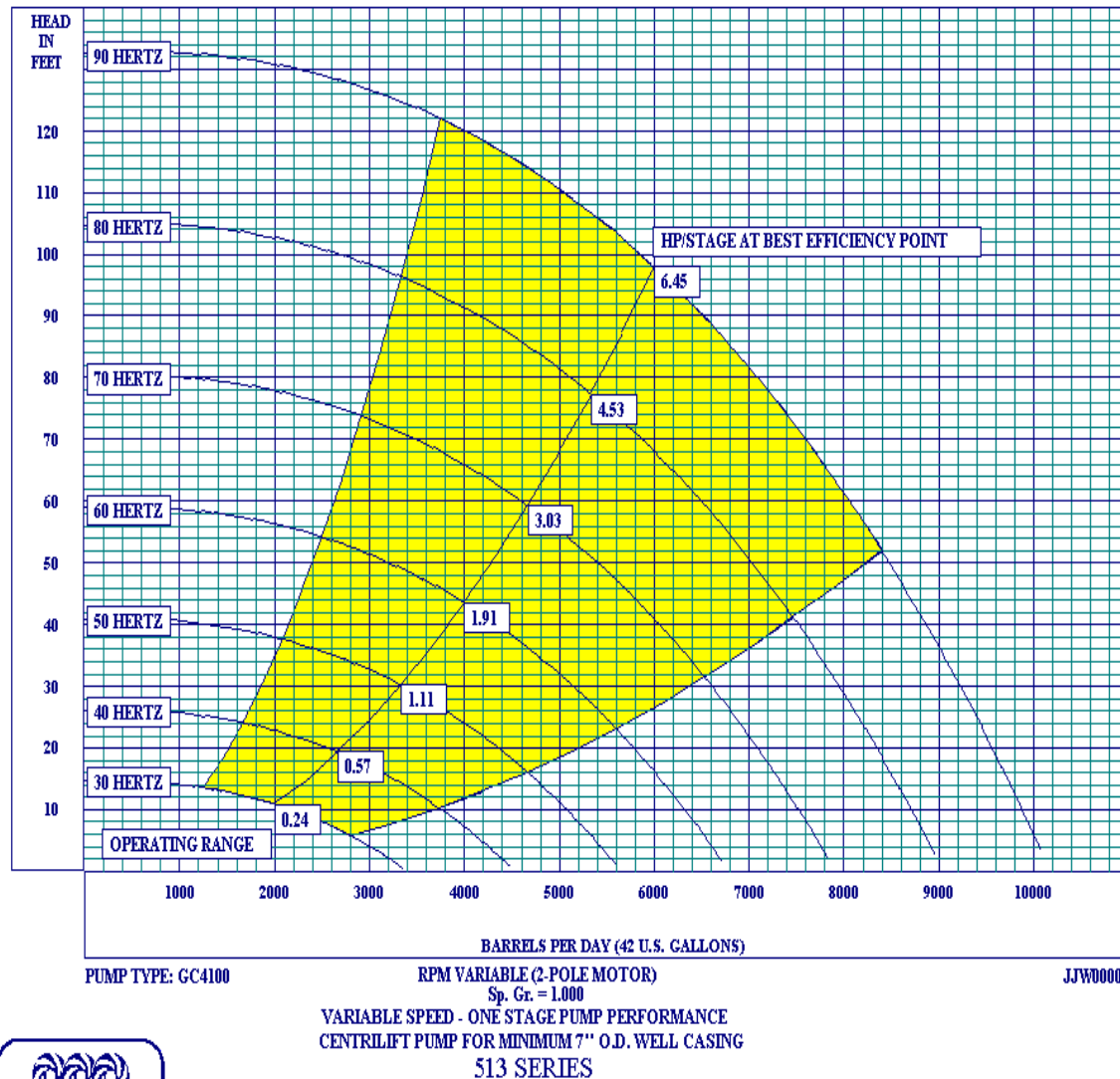
Downhole Centrifugal Pump

- May be driven at other speeds either due to power source or intentionally to modify performance
 - Flow = f (rpm) = f (frequency)
 - Head = f (rpm²) = f (frequency²)
 - Power = f (rpm³) = f (frequency³)



Downhole Centrifugal Pump

If frequency jumped from
60 Hz to 80 Hz how much
Power increment %
Head increment %



1 HP = 745 watt



GC4100

VSD Problem

The formula for efficiency of an ESP stage is :

$$\eta = \frac{0.00000736(Q, bpd)(Head, ft)(\gamma_{liquid})}{BHP}$$

(A) If a pump stage is brought from 60 Hz to 80 Hz, how does the efficiency change?

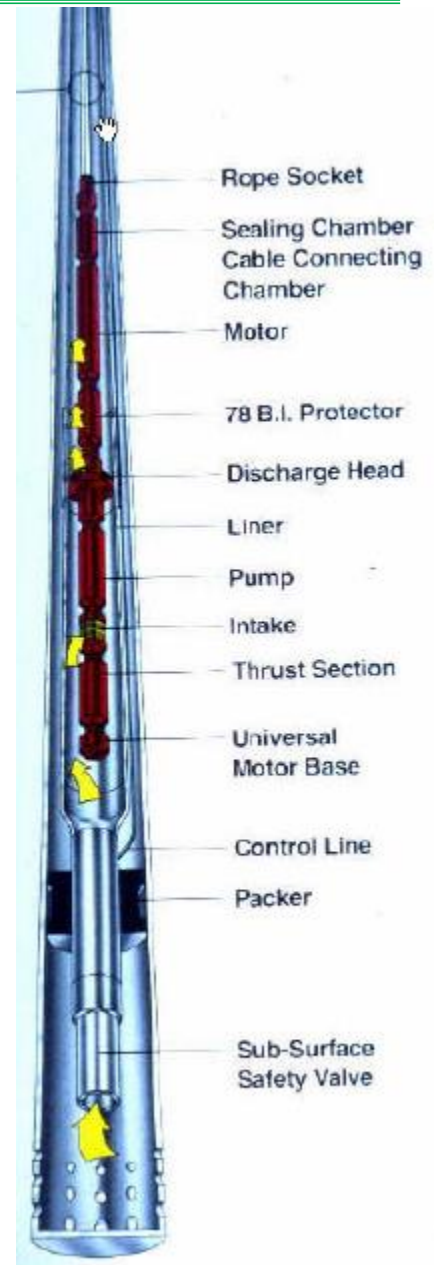
VSD variable speed drive

Introduction to ESP

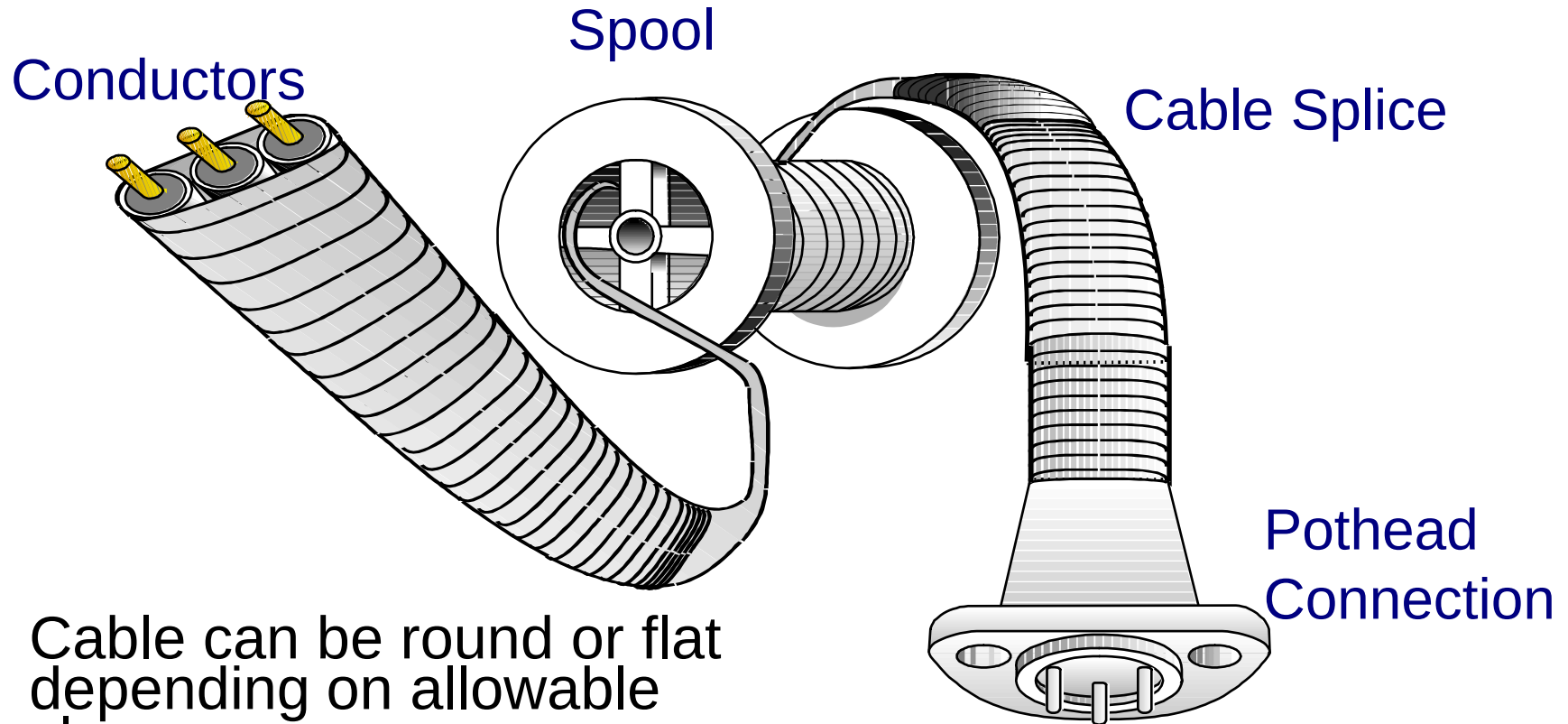
Cable Suspended ESP System

Cable Suspended ESP:

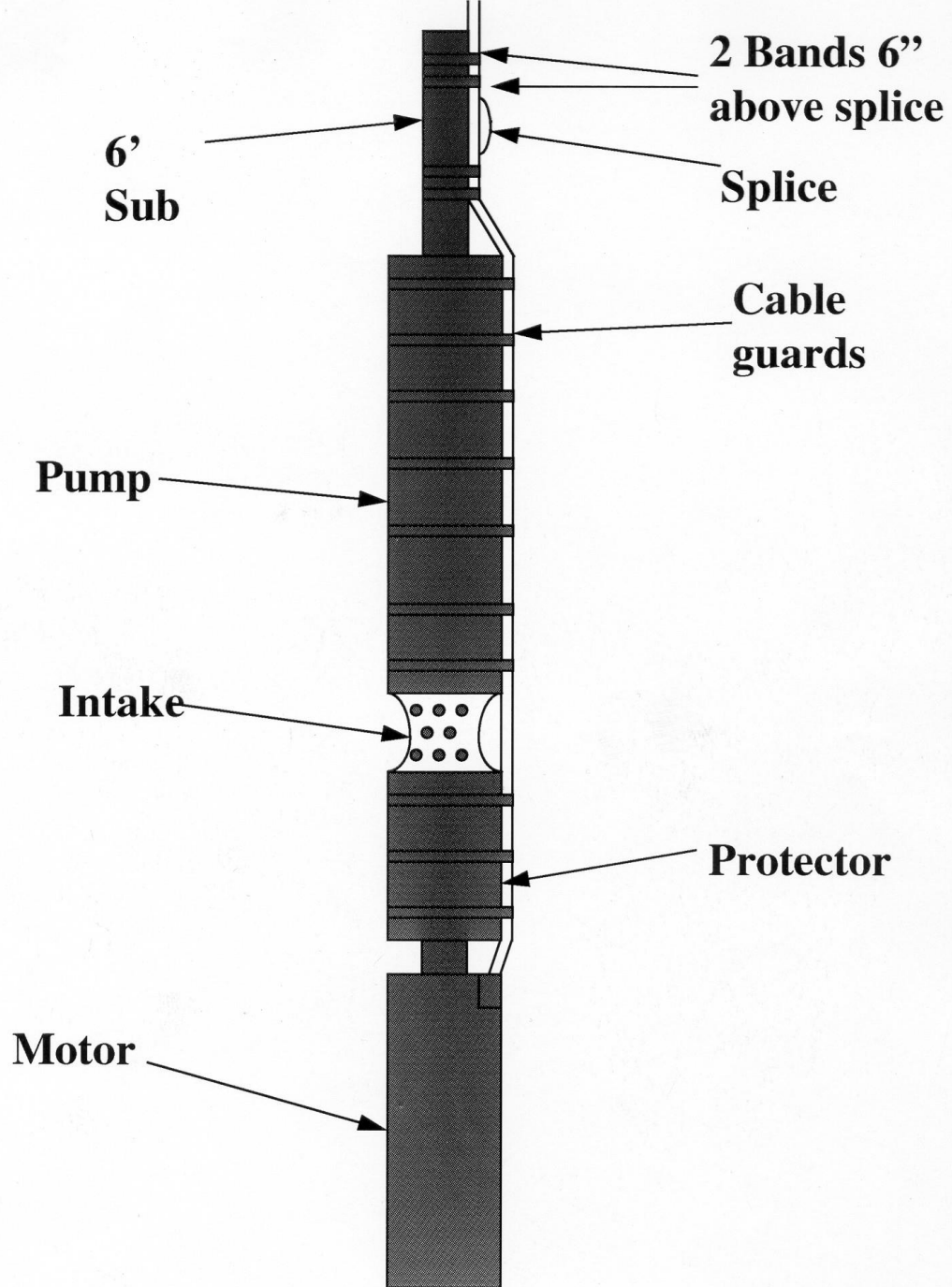
- The unit is lowered in the well without using a tubing. It is suspended from a cable and the power cable is banded to it.
- A special seating element supports the pump and provides locking to avoid excessive torque on the cable.
- Differently from the conventional installations, the motor is located above the pump.
- The system produces through the annular.
- Its main advantage is the reduction in all costs associated with tubing pulling job, specially offshore



Cable



- Cable can be round or flat depending on allowable clearance
- Upper and lower pigtails can be spliced on to allow for packers and wellhead connections



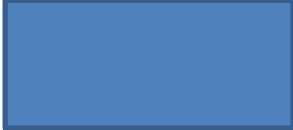
BANDING AT /NEAR PUMPING UNIT

Band 4' on either
side of coupling: or
if low profile
tubing or new
upset tubing, band
2-3 ' on each side
of collar

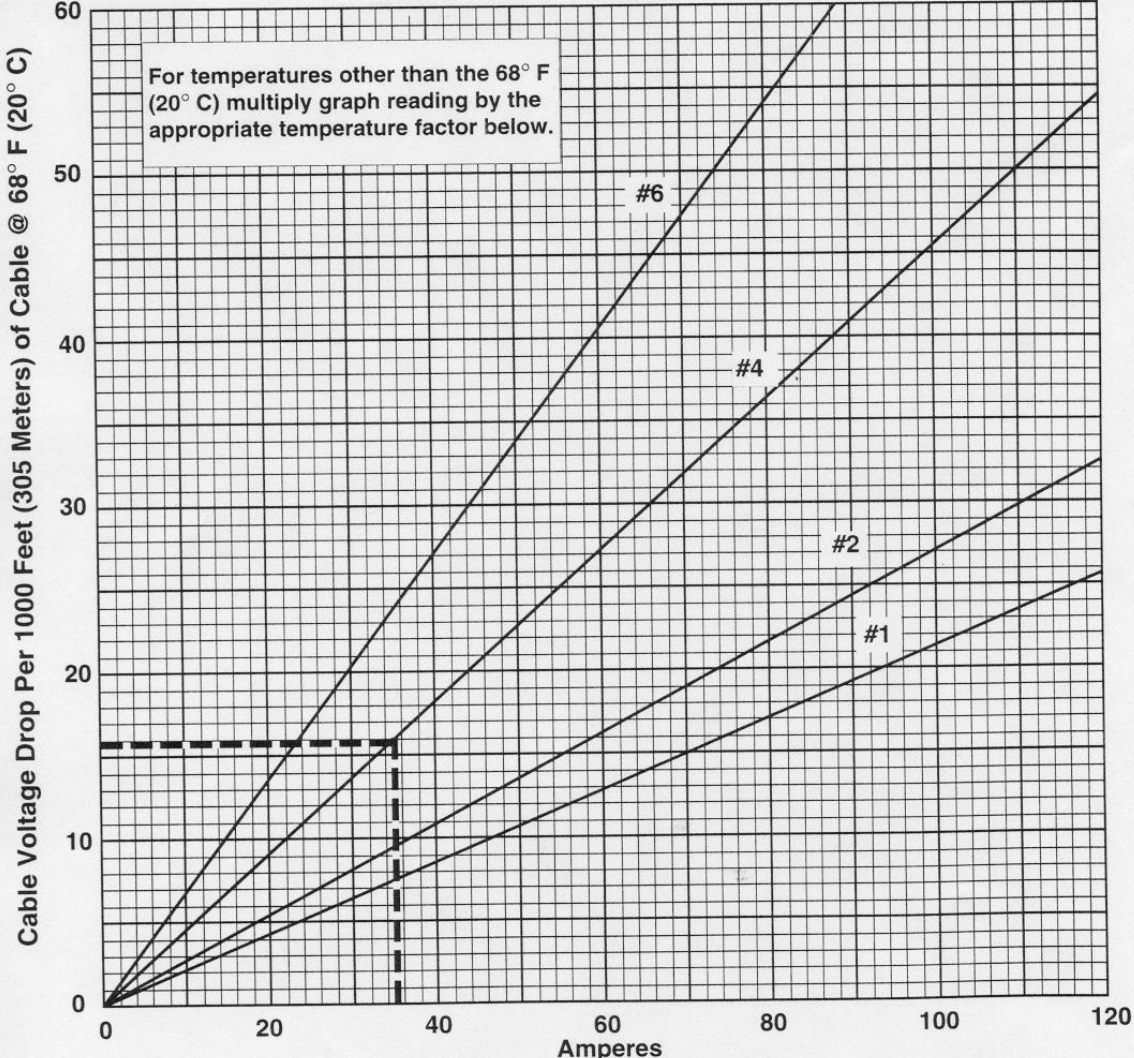
Cable
encapsulating
guards are
available for
collar locations

No bands in center of
tubing

BANDING ON TUBING

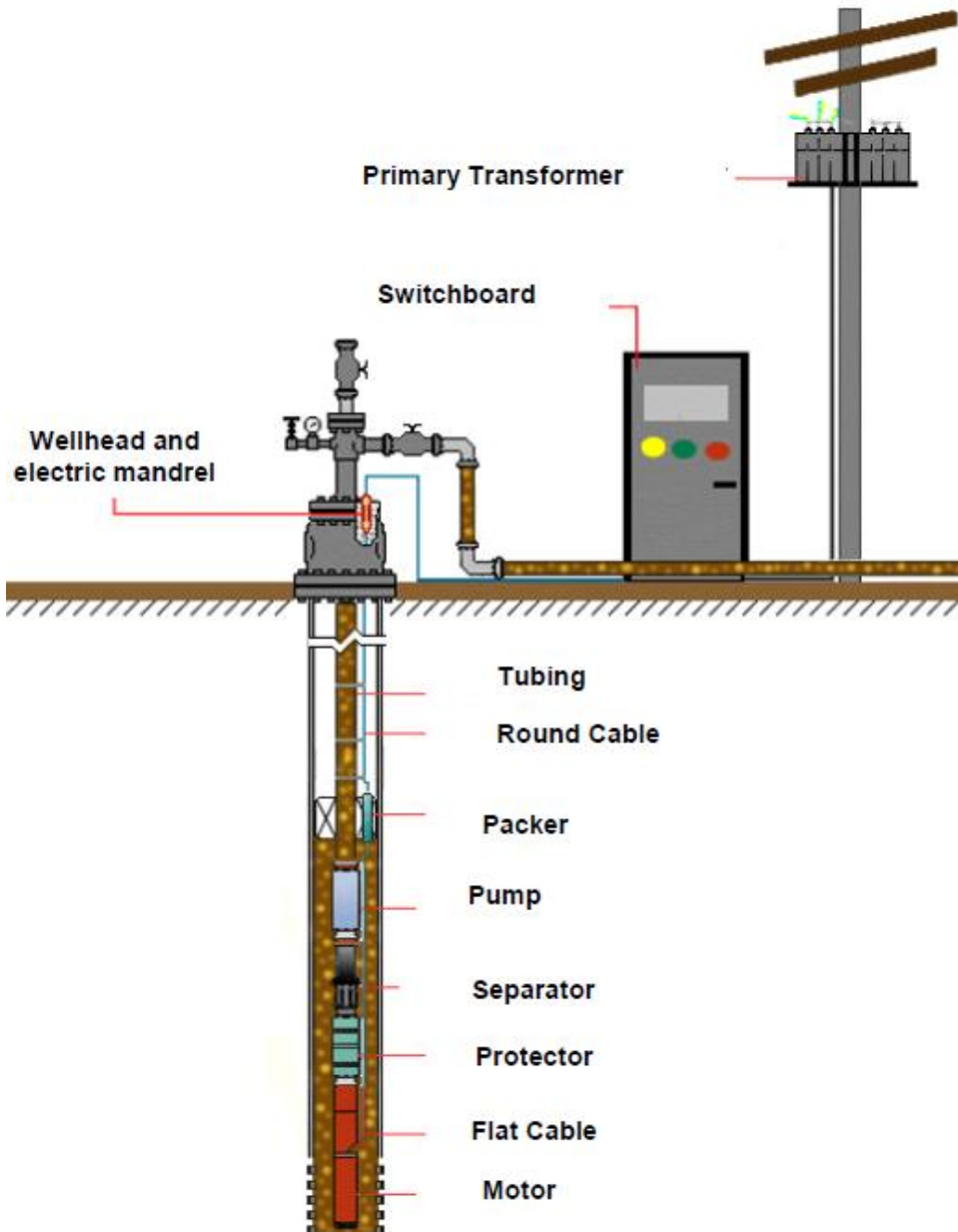


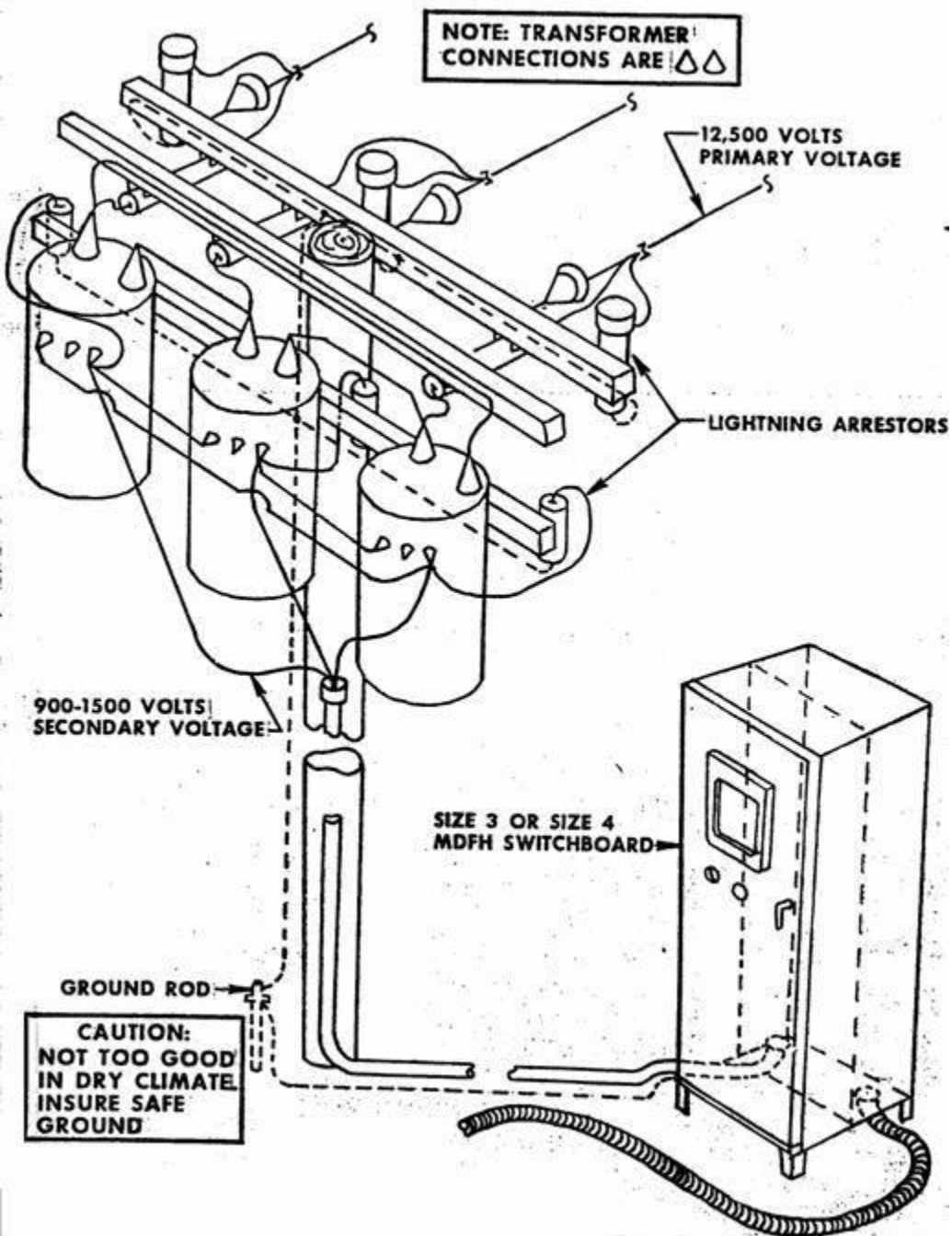
Cable voltage drop vs. temperature and depth for various sizes of cable



Temp. °F (°C)	Multiplying Factor	Temp. °F (°C)	Multiplying Factor	Temp. °F (°C)	Multiplying Factor
100 (38)	1.070	170 (77)	1.223	240 (116)	1.376
110 (43)	1.092	180 (82)	1.245	250 (121)	1.398
120 (49)	1.114	190 (88)	1.267	260 (127)	1.420
130 (54)	1.136	200 (93)	1.288	270 (132)	1.441
140 (60)	1.157	210 (99)	1.310	280 (138)	1.463
150 (66)	1.179	220 (104)	1.332	290 (143)	1.485
160 (71)	1.201	230 (110)	1.354	300 (159)	1.507

ESP Components



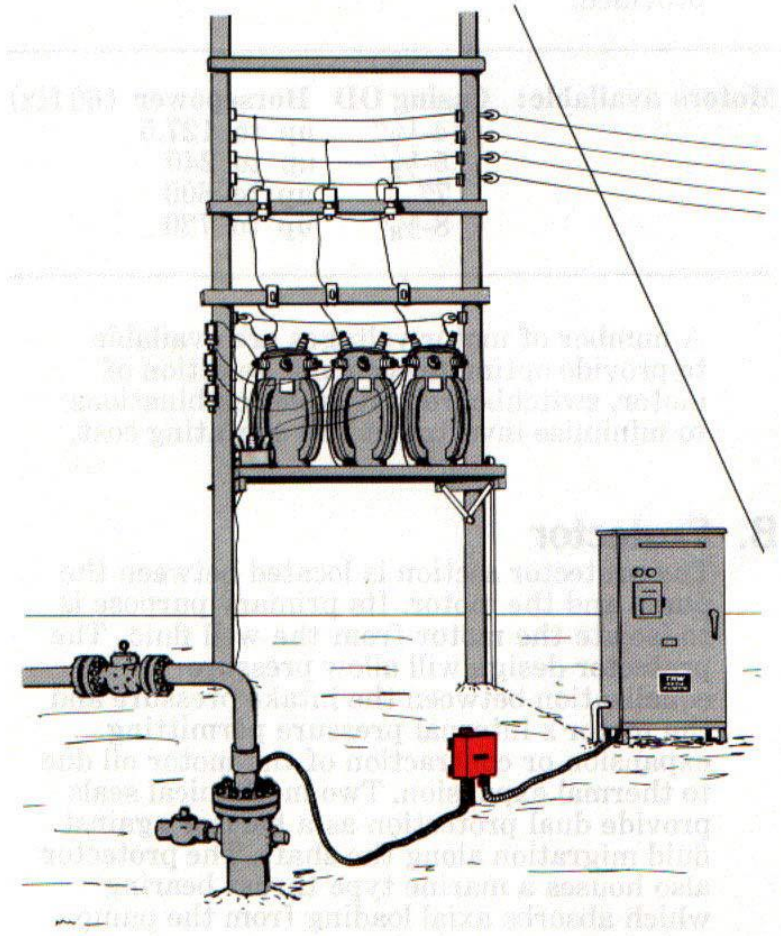


**TRANSFORMER 12,500
TO 1000 VOLTS AS
FOR VSD 460 VOLTS**

Introduction to ESP

Surface Components - Transformer

transformer system is used to *step-up* or *step-down* the voltage from the primary line to the motor of the submersible pump. Because a range of operating voltages may be used for submersible pump motors, the transformer must be compatible with the selection of the motor voltage.



Introduction to ESP

Surface Components - Switchboard

The switchboard controls the pump motor and provides overload and underload protection.

Protection against overload is needed to keep the motor windings from burning.

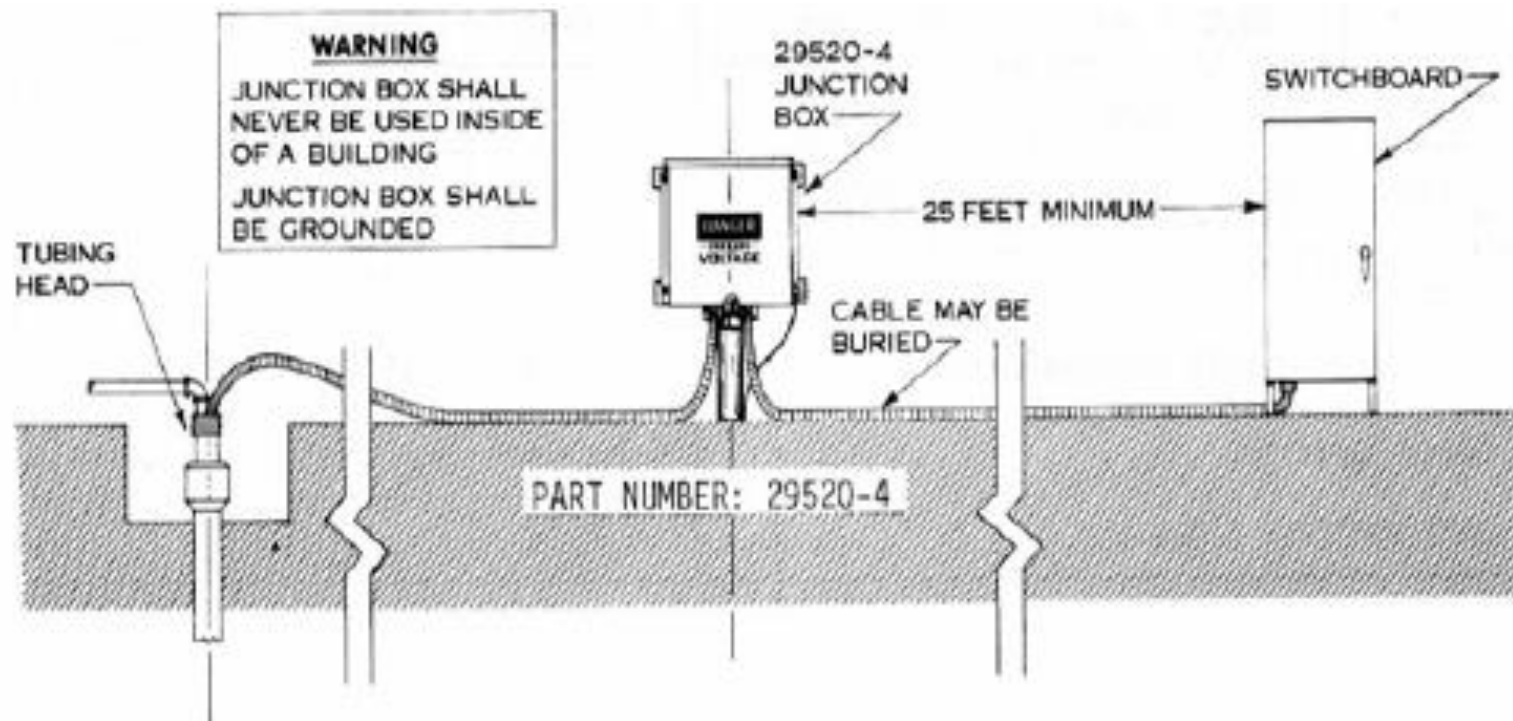
Protection during underload is needed because low fluid flow rates will prevent adequate cooling of the motor.



Introduction to ESP

Surface Components – Junction Box

The junction box connects the power cable from the switchboard to the power cable from the well. It provides an explosion-free vent to the atmosphere for any gas that might migrate up the power cable from the wellbore.



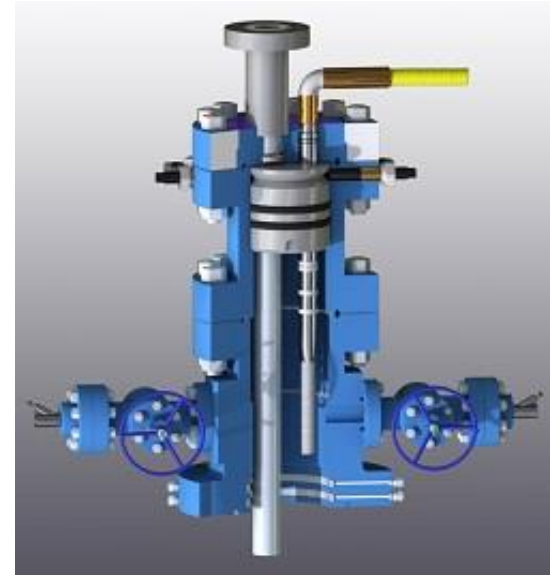
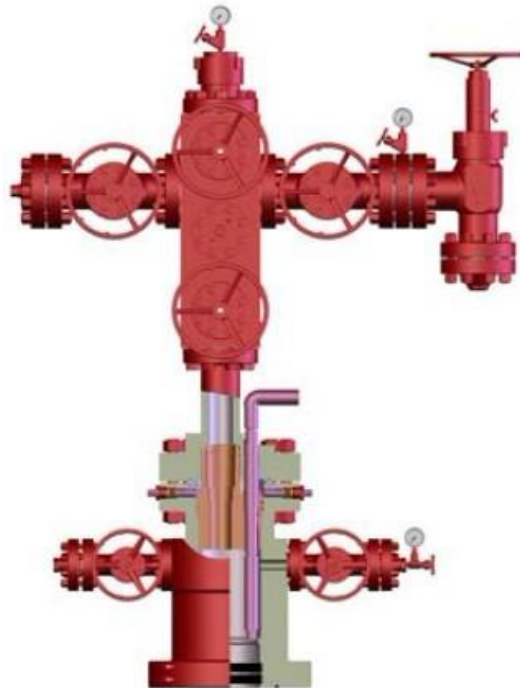
Introduction to ESP

Surface Components – Wellhead

Must provide means for installing the cable with adequate seal

May include adjustable chokes, bleeding valves

Onshore wellheads have a rubber seal and offshore have a electric mandrel

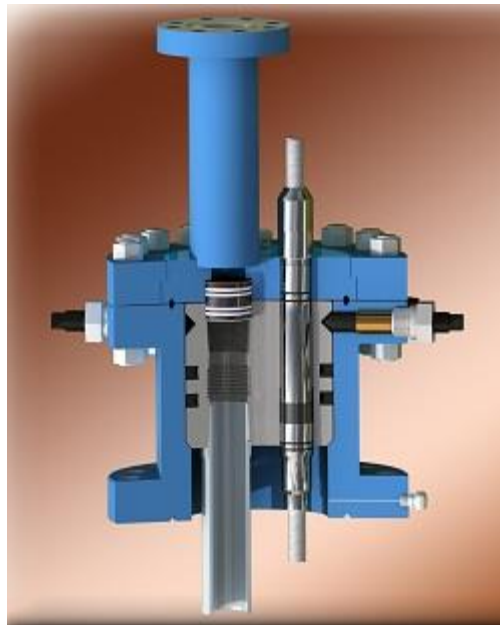


Introduction to ESP

Surface Components – Wellhead

The Safe-T-Lok is supplied with factory molded cable on both the top and the bottom. The lower cable will be spliced to the ESP cable, and the top cable will be connected to the junction box.

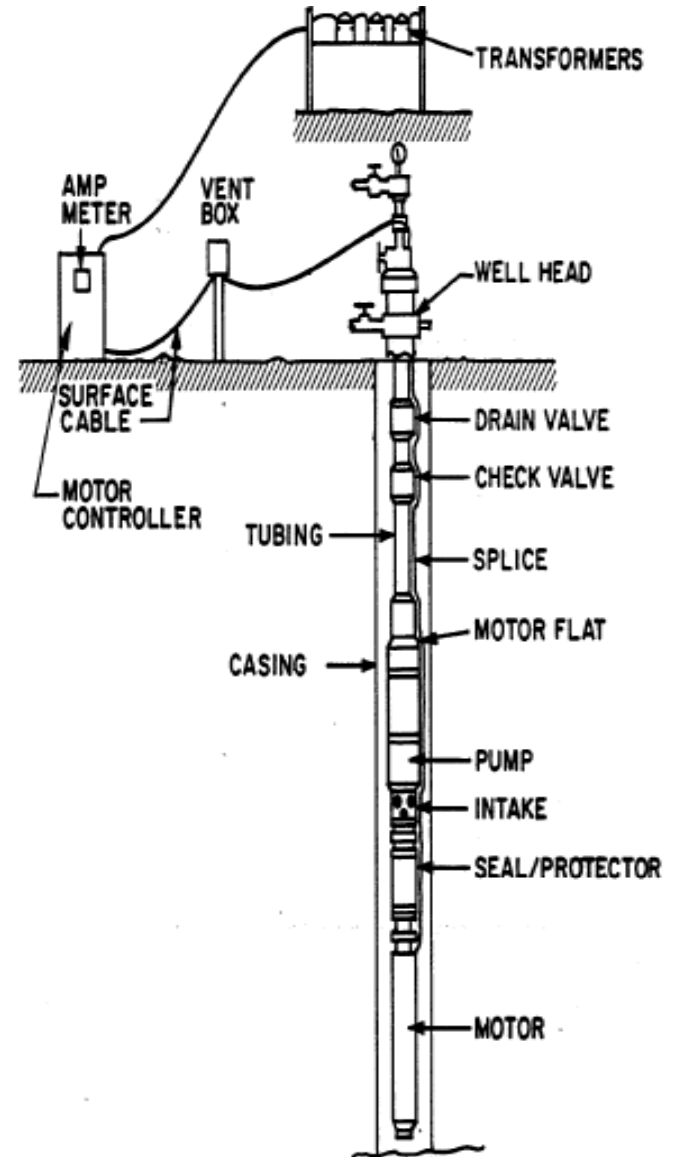
The Safe-T-Lok is installed in the wellhead by feeding through the tubing hanger from below



Introduction to ESP

Subsurface Components – Check Valve

A check valve is installed about two to three joints above the ESP pump to maintain a full liquid column in the tubing string during equipment shut down periods. It prevents leaking of the fluid from the tubing down through the pump when the pump is not running.



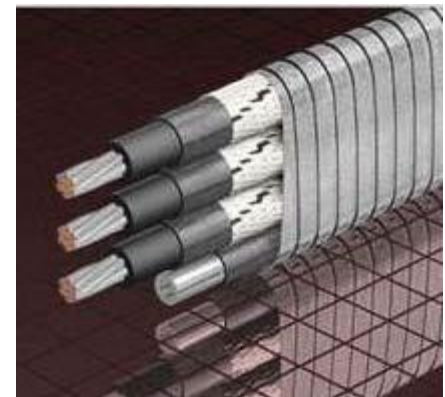
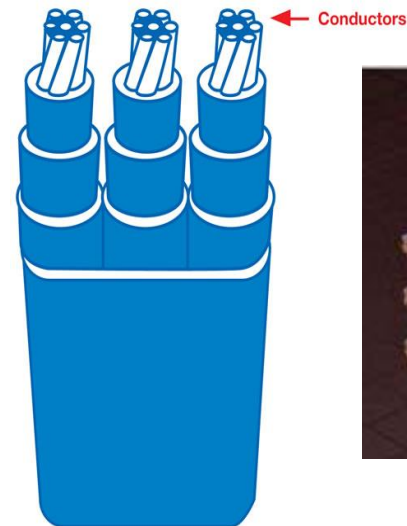
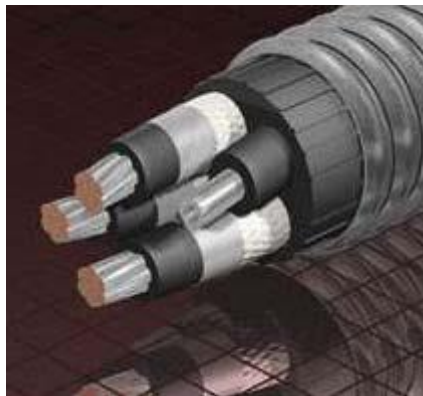
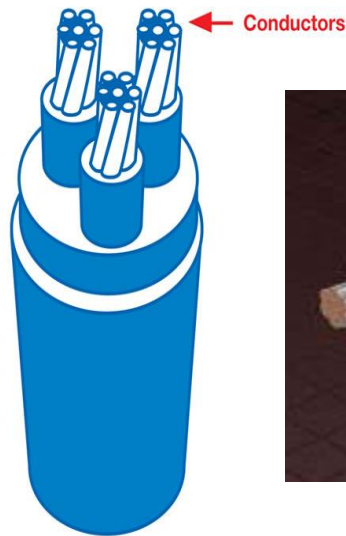
Electrical Submersible Pump

Introduction to ESP

Subsurface Components – Electric Cable

A power cable runs from the junction box then through the wellhead and all the way to the bottom to supply power to the pump motor.

Cable is available in round and flat styles

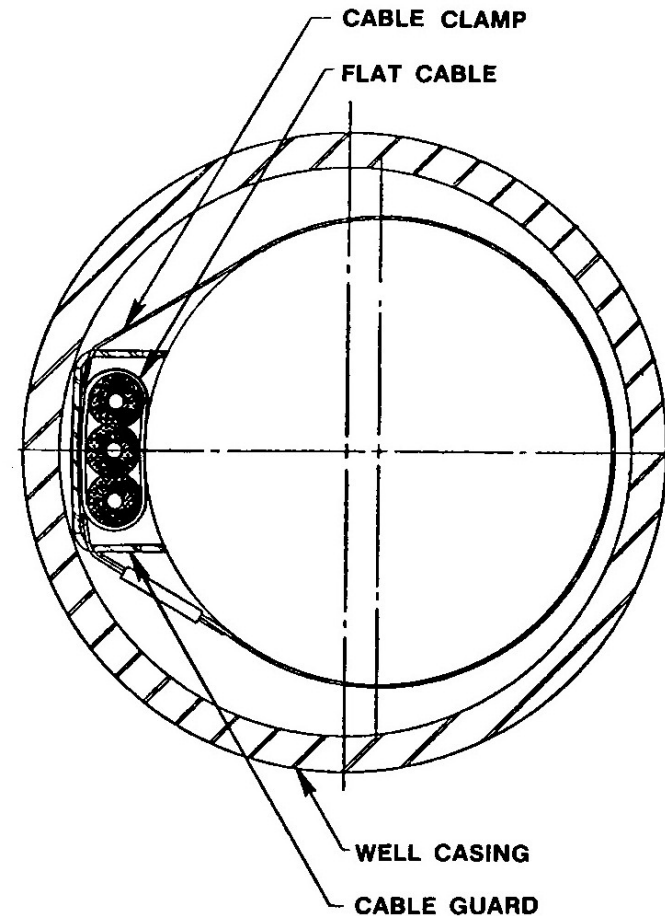


Introduction to ESP

Subsurface Components – Cable Protection

Cable Guards: Used to protect the motor lead cable avoiding the direct contact of the cable with the casing standard. Standard length 8 ft.

Cable clamps: used to tie the cable to the tubing.

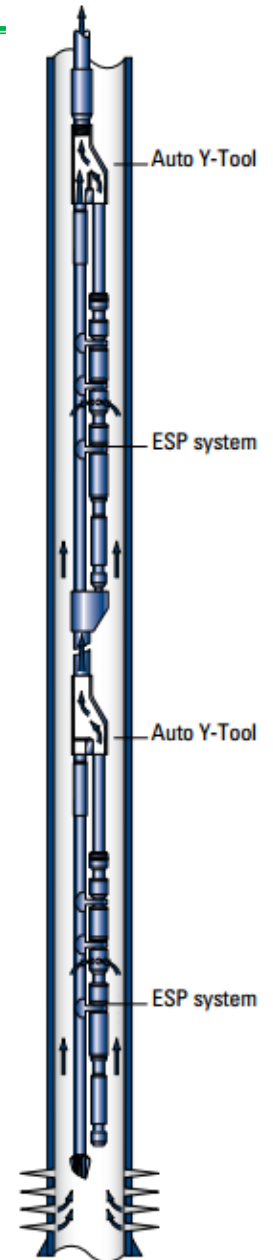
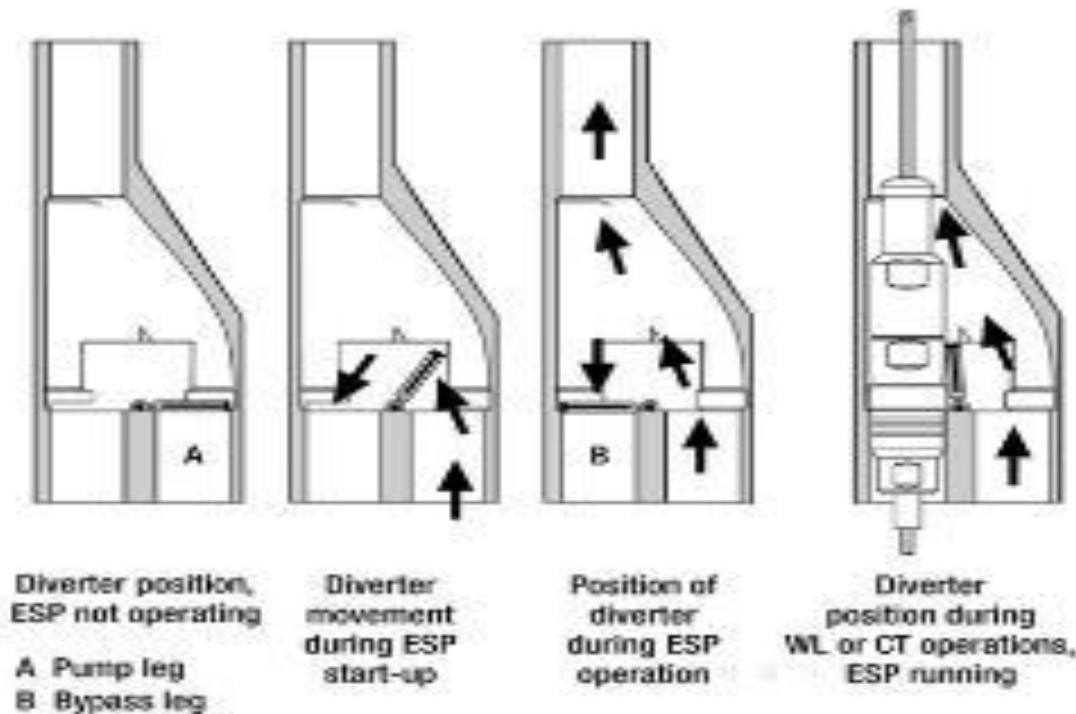


ESP

Subsurface Components –ESP Bypass System

Wireline or coiled tubing plugs can be supplied to seat in a nipple profile in the Y-tool to enable intervention or logging operations without retrieval of the completion

Can be also used for installing two parallel ESPs in the well.



ESP

Subsurface Components – Electric Cable

The proper selection of the cable and the conductors depends on:

- a) The expected amperage that will flow through the cable to the motor**
- b) The calculated voltage drop in the line from the surface to the pump.**
- c) The space that exists between the tubing collar and the casing (even though the cable is banded to the tubing at selected points, there must be enough space to install and pull the pump without damaging the cable or hanging it in the well).**
- d) The equipment operating environment - such as the operating pressure and temperature at pump depth.**

Introduction to ESP

Subsurface Components – Cable Amperage

The first consideration in selecting cables is amperage. The limits on amperage for cables containing copper conductors are as follows:

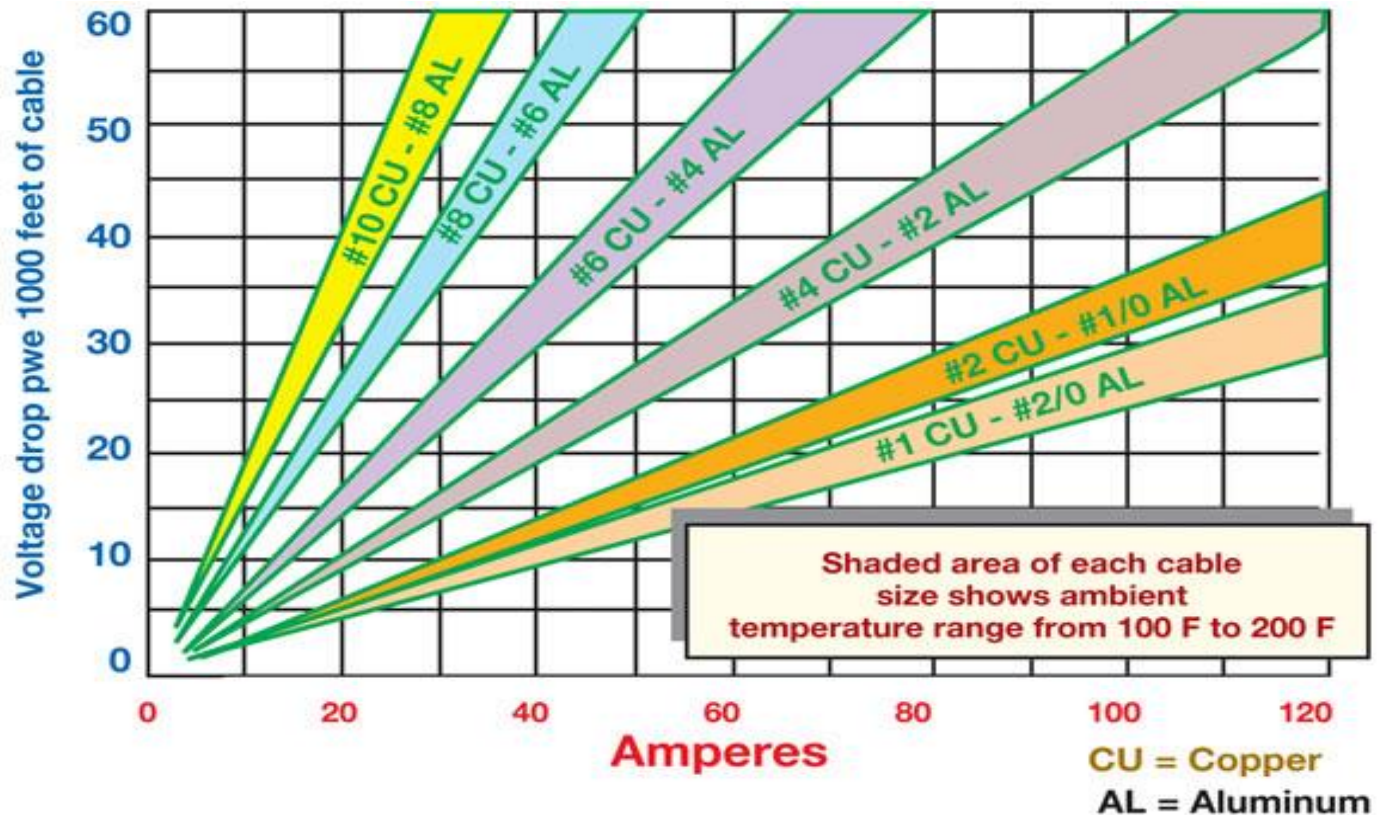
Cable No.	Maximum Amperage
1	115
2	95
4	70
6	55

Note that the cable with the smaller number has the larger diameter. Thus, a Number 1 cable can carry a maximum of 115 amps.

ESP

Subsurface Components – Voltage Drop

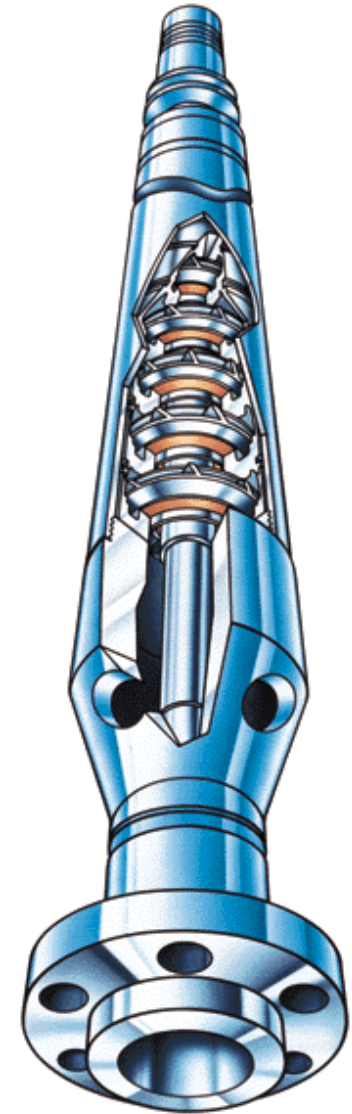
The second selection consideration is the voltage drop that will occur between the wellhead and the pump. Normally, the maximum voltage drop for an electrical cable is about 30V per 1000 feet.



Introduction to ESP

ESP Operating Principles

- a) Electrical submersible pumps are multi-staged centrifugal pumps. Each stage consists of a rotating impeller and a stationary diffuser.
- b) The performance of the pump depends on the stage design, size, rotational speed, and fluid being pumped.
- c) The rotating movement of the motor is transferred through the shaft to the impeller.
- d) The overall length of a single pump section is limited to 25-30 feet to facilitate assembly, transportation, and handling.



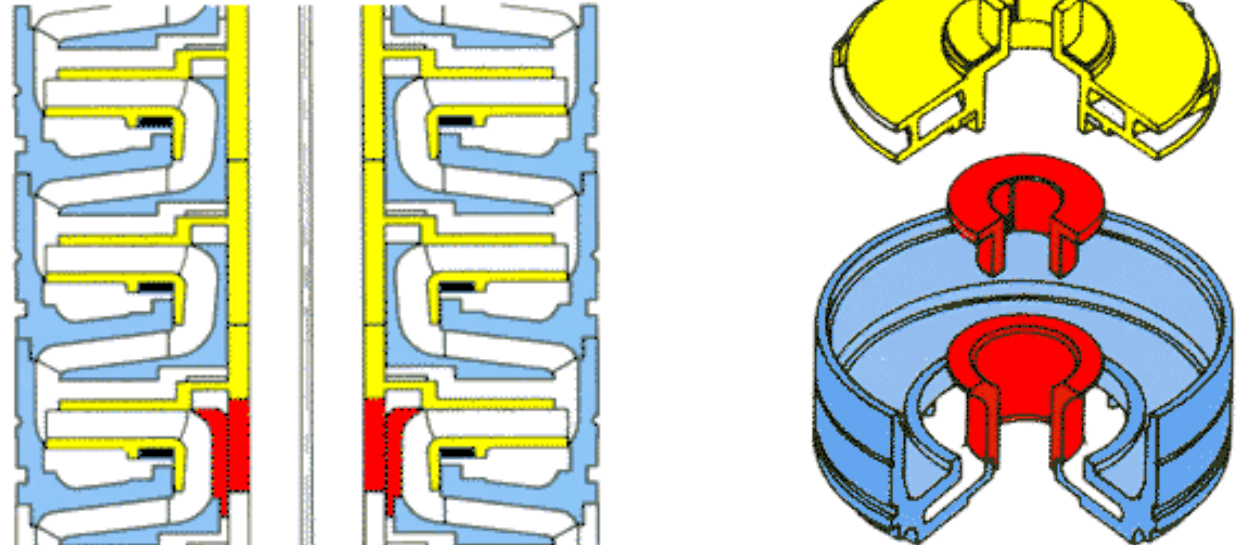
Introduction to ESP

ESP Operating Principles

Each stage consists of an impeller and a diffuser

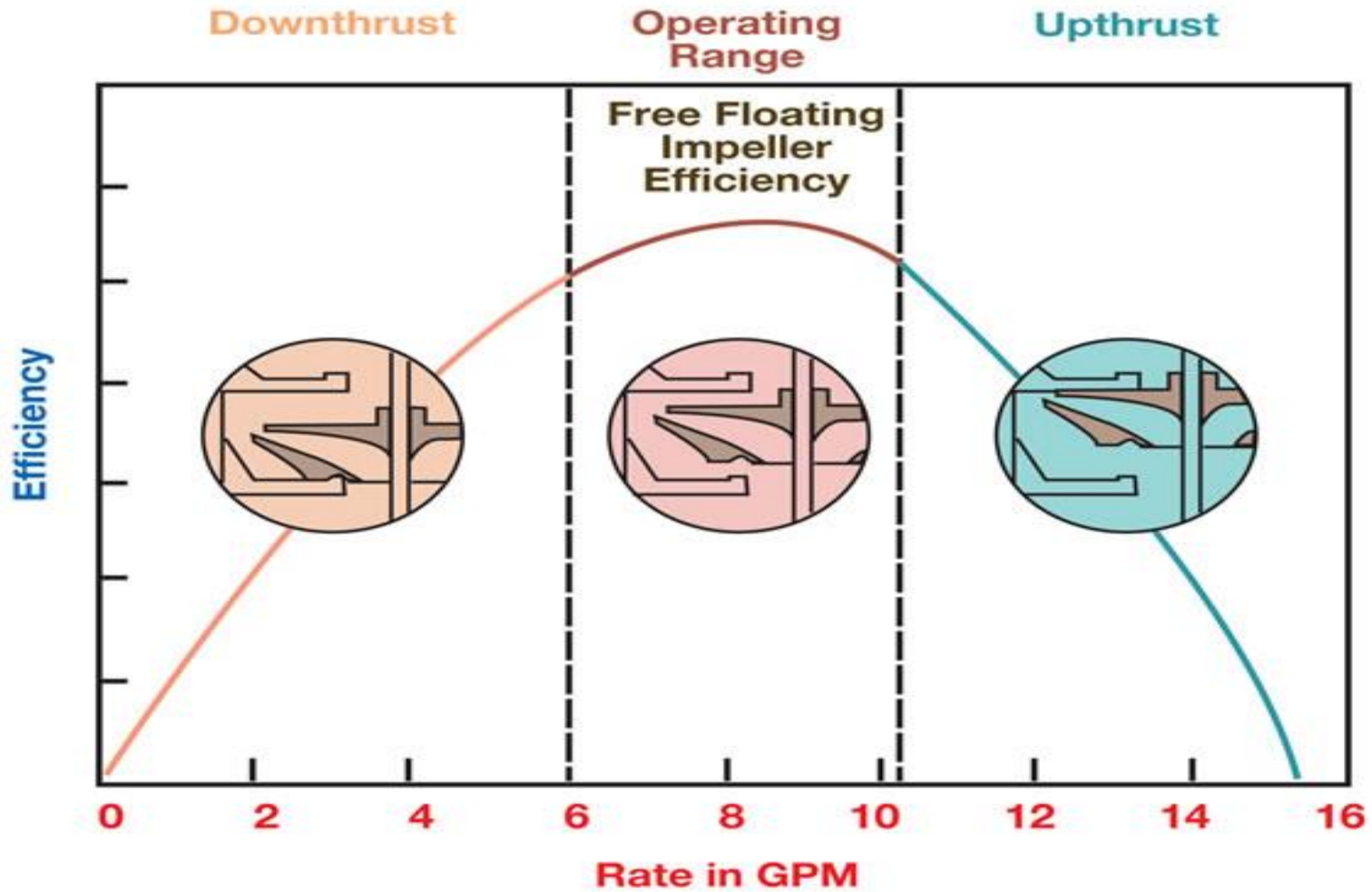
The rotating impeller takes the fluids and imparts kinetic energy from the rotating shaft to the fluids

The stationary diffuser converts the kinetic energy of the fluids into pressure



ESP

ESP Operating Principles



Introduction to ESP

ESP Operating Principles

A pump's impellers are designed to operate efficiently over a specific capacity range. Operating the pump below its design capacity causes the impeller to *downthrust* against the diffuser, resulting in wear on the bearings and washers.

Conversely, if the pump operates above its design capacity, the impeller *upthrusts* against the upper part of the diffuser, causing similar wear.

Ideally, the impeller should float freely, and will do so throughout its recommended operating range. This recommended operating range will allow the pump to run at highest efficiency

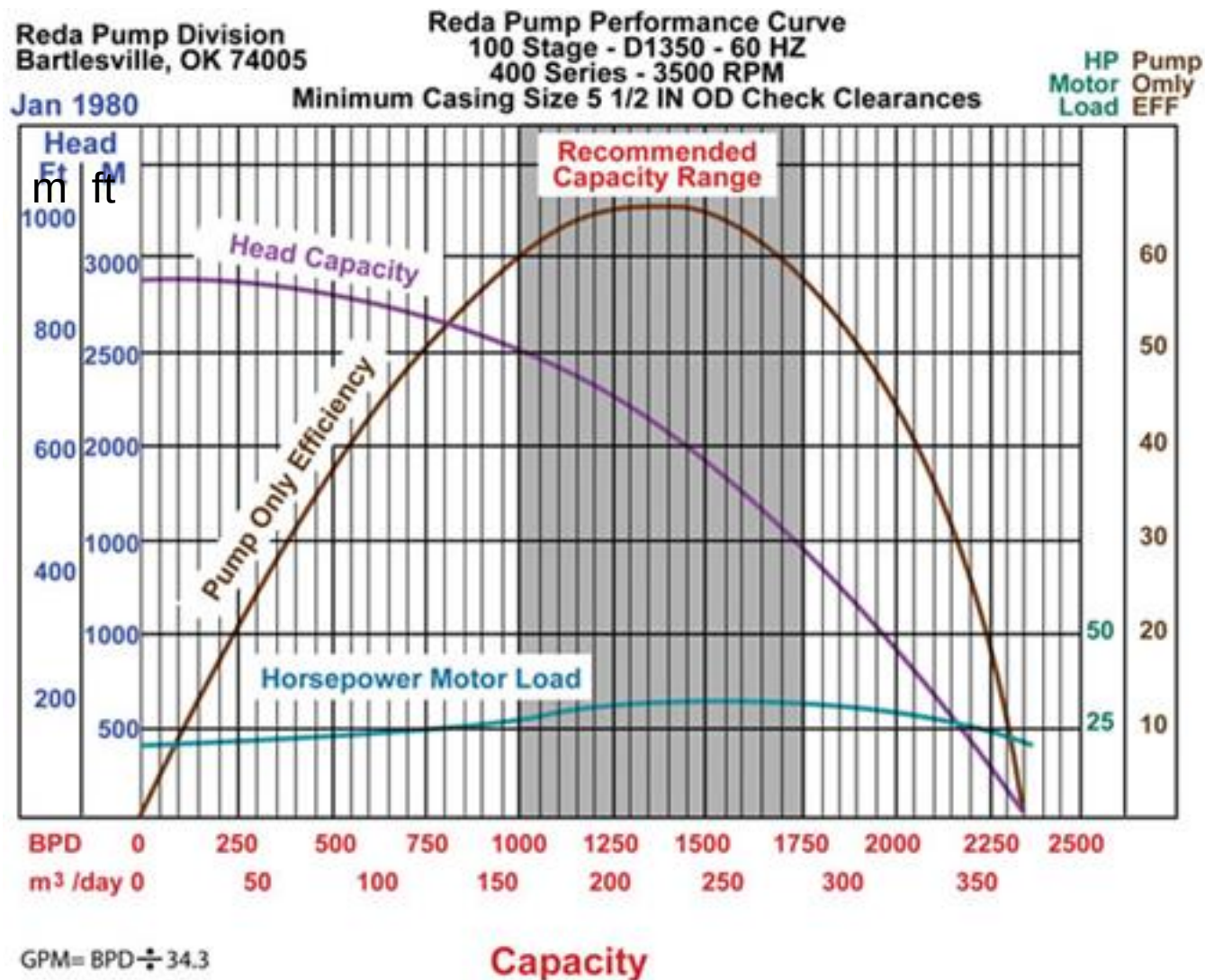
ESP

ESP Operating Principles

O.D. of casing	Casing weight	Motor O.D.	Pump O.D.
4.50 inch	11.5 lb/ft	3.75 inch	3.375 inch
	9.5 lb/ft		
5 inch	All weights	3.75 inch	3.375 inch
5.5 inch	20 lb/ft	4.5 inch	4.0 inch
	17 lb/ft		
	15.5 lb/ft		
7 inch	28 lb/ft	4.5 inch	4.0 inch
	26 lb/ft	5.5 inch	5.375 inch
	24 lb/ft		
	20 lb/ft		
8.625 inch or greater	All weights	4.5 inch	4.0 inch
		5.5 inch	5.125 inch
		7.375 inch	6.75 inch

Introduction to ESP

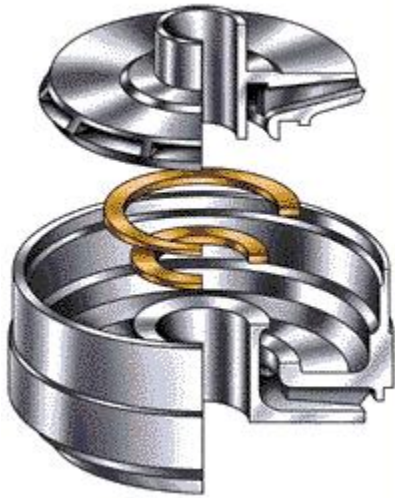
ESP Operating Principles



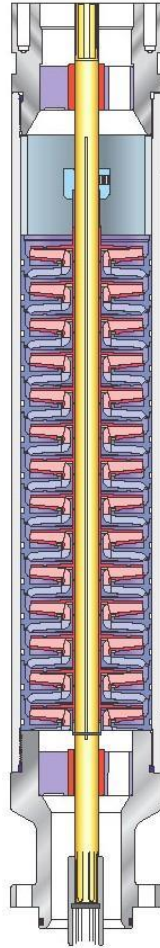
Introduction to ESP

ESP Classification

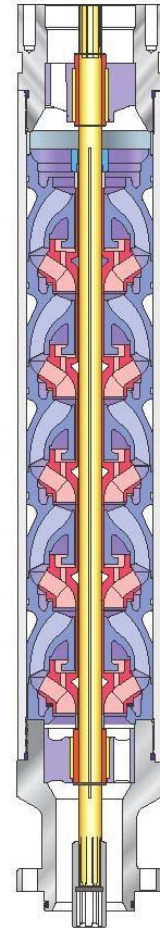
ESPs can be classified into two main categories: Radial flow and Mixed flow



Radial Flow Pump



Mixed Flow Pump



Introduction to ESP

Subsurface Components – Gas Separator



Gas Separator

- Used in applications where free gas causes interference with pump performance
- Units separate some of free gas from fluid stream entering pump to improve pump's performance



Introduction to ESP

Subsurface Components – Gas Separator

Separates the free gas in order to reduce the quantity of gas that flows into the pump.

There are two types: static and rotary gas separator.

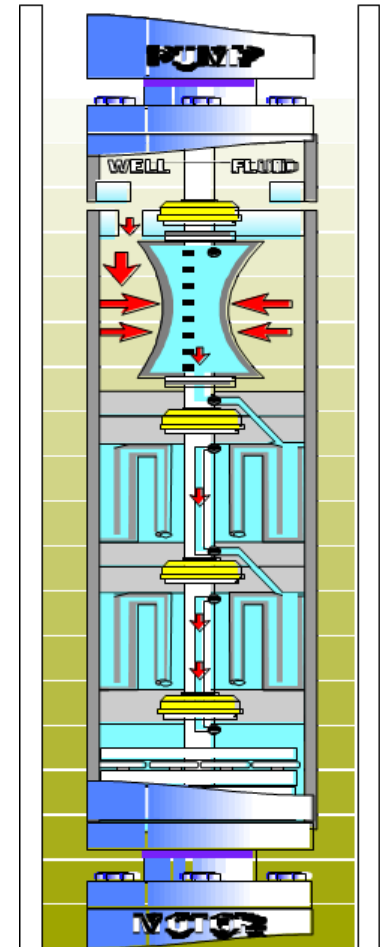
Static: No applying any additional mechanical force. They provide a tortuous path that turns the fluid stream and moves it down toward the inlet ports. Some of the free gas accompanies the liquid to the intake and a portion is separated.

Dynamic gas separators, on the other hand, actually impart energy to the fluid to separate the vapor from the fluid...

-

Seal Section

- Seal equalizes pressure and prevents well fluid from entering motor by using:
 - Mechanical seals
 - Bladder system
 - Labyrinth system



Introduction to ESP

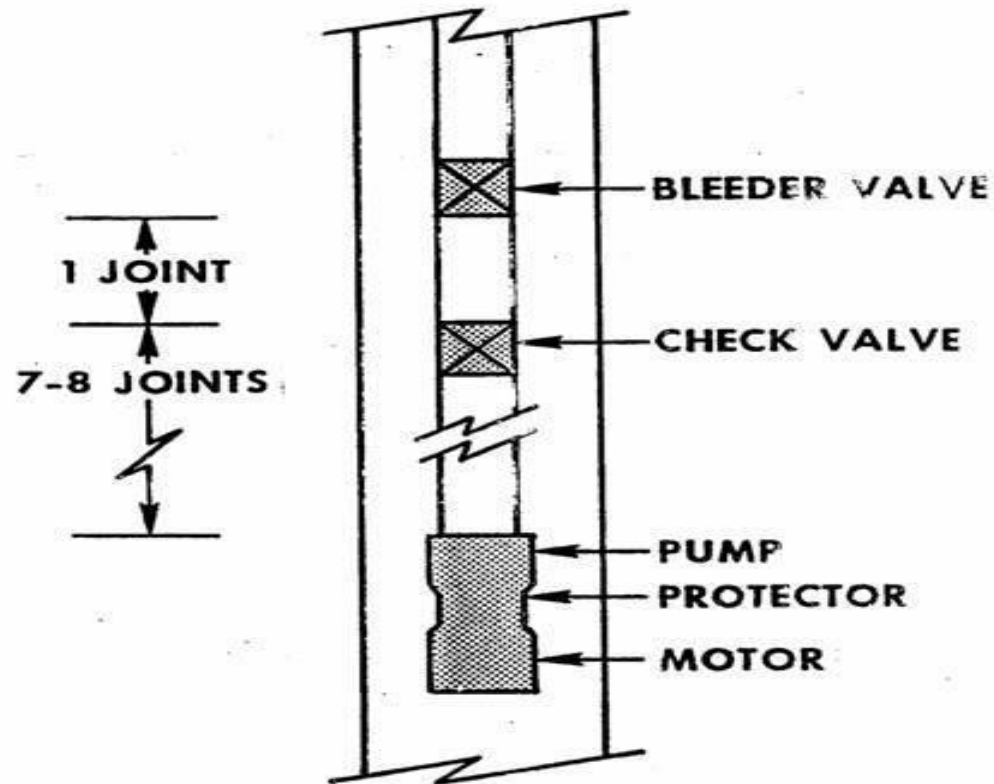
Subsurface Components – Protector or Seal

Serves as the connection between the motor shaft and the pump shaft

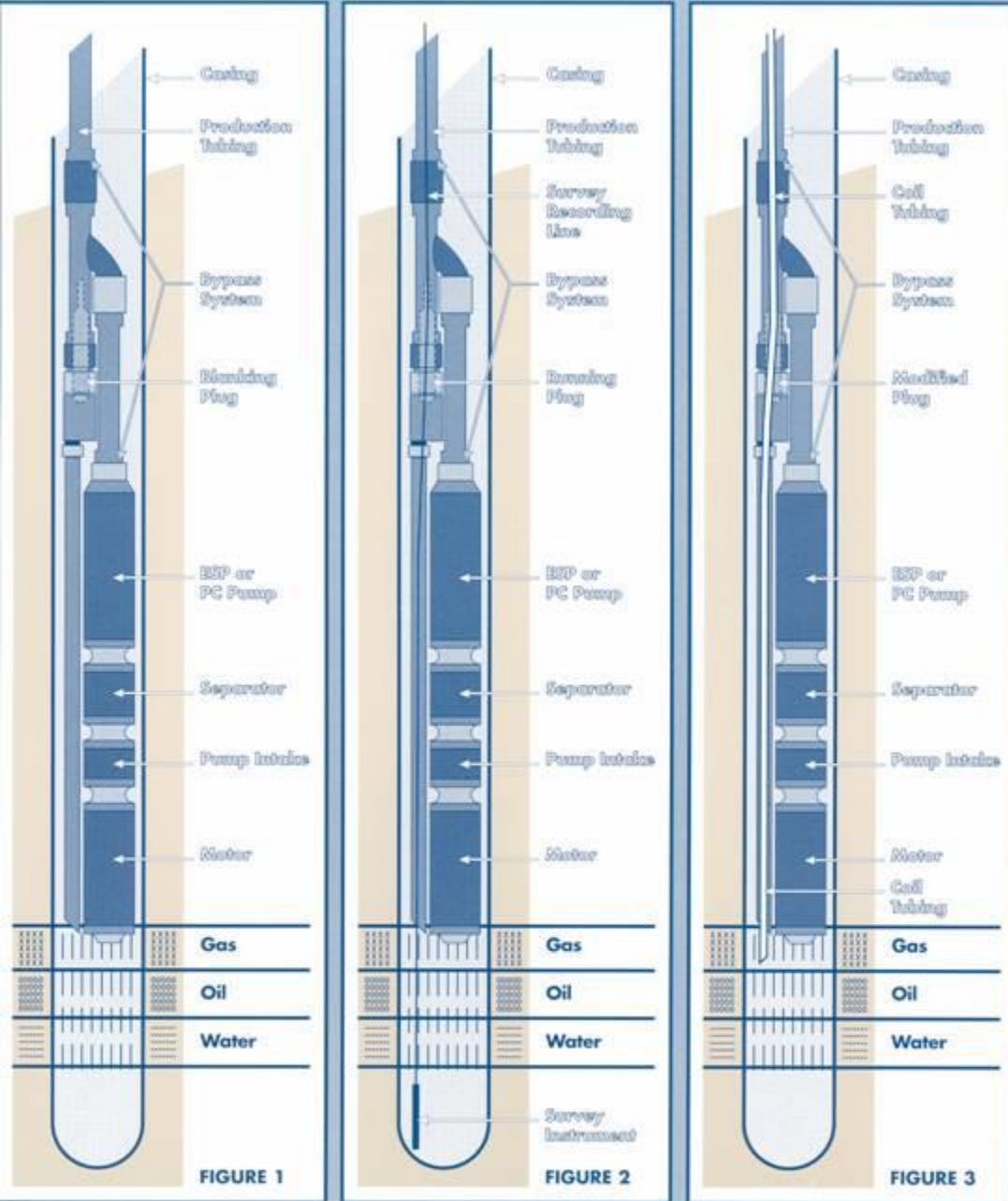
- Prevents the entry of well fluid into the motor
- Provides an oil reservoir to compensate for expansion and contraction of motor oil
- Support the axial thrust developed by the pump on the seal thrust bearing
- Pressure equalizer
- Use multiple redundant barrier chambers isolate the fluidsto



ESP



- NOTES:**
- (1) FOR WATER WELLS THE CHECK VALVE MAY BE ONLY 1 JOINT FROM PUMP
 - (2) FOR GASSY OIL WELLS THE CHECK VALVE SHOULD BE 7-8 JOINTS ABOVE PUMP
 - (3) THE CHECK VALVE CAN BE WIRELINE RETRIEVABLE IN WHICH CASE A BLEEDER VALVE IS NOT USED

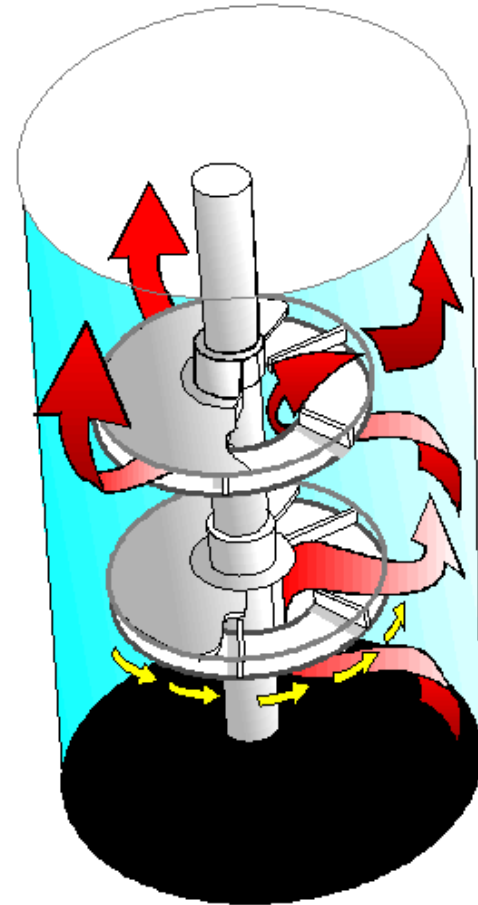


The Y tool is commonly used offshore to allow well work below the ESP, without pulling the ESP.

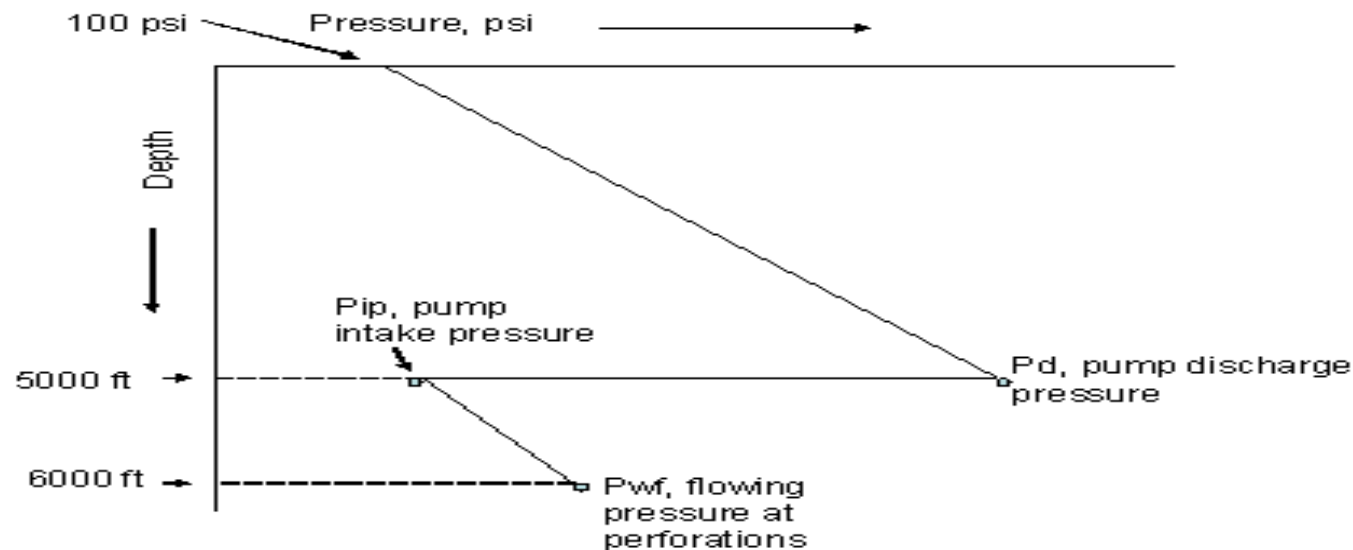
The **OSTS By-Pass Tool** facilitates instrument readings by being placed into operation without interrupting pump activity. The **By-Pass Tool** allows instruments to be run inside the tubing string and be placed just below the pump. **OSTS By-Pass Tools** have been designed and engineered to be available in Standard 2-3/8", 2-7/8" and 3-1/2" API Tubing, Casing sizes, and weights. See the reference table on the back page.

Centrifugal Pump Stage

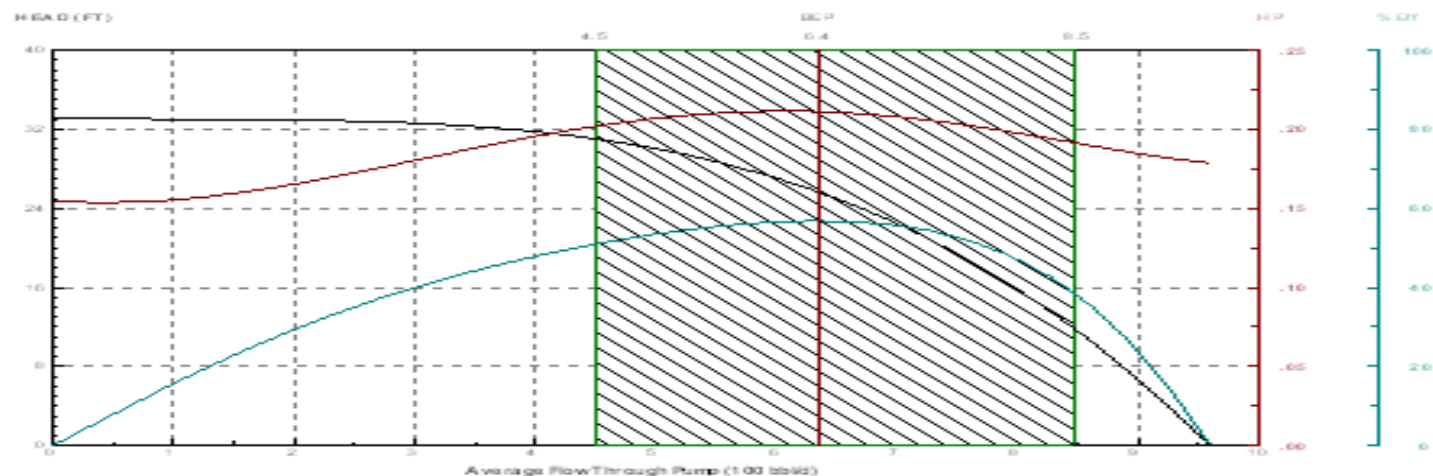
- Works by transferring energy provided by driver to fluid to be moved
- Rotating impeller uses centrifugal force it generates to increase velocity (kinetic energy) of fluid
- Stationary diffuser diffuses, or reduces, velocity of fluid -- converting kinetic energy into potential energy or pressure



The flowing gradient above the pump and below the pump is $.433 \times \text{SG}$ psi/ft. The SG is 0.9 for the fluid. The well is producing 650 bpd. The PI of the well is 1.0 bpd/psi. The shut-in pressure of the well is 2000 psi. How many stages of the FC 650 pump are in the pump if it is producing according to the head curve?

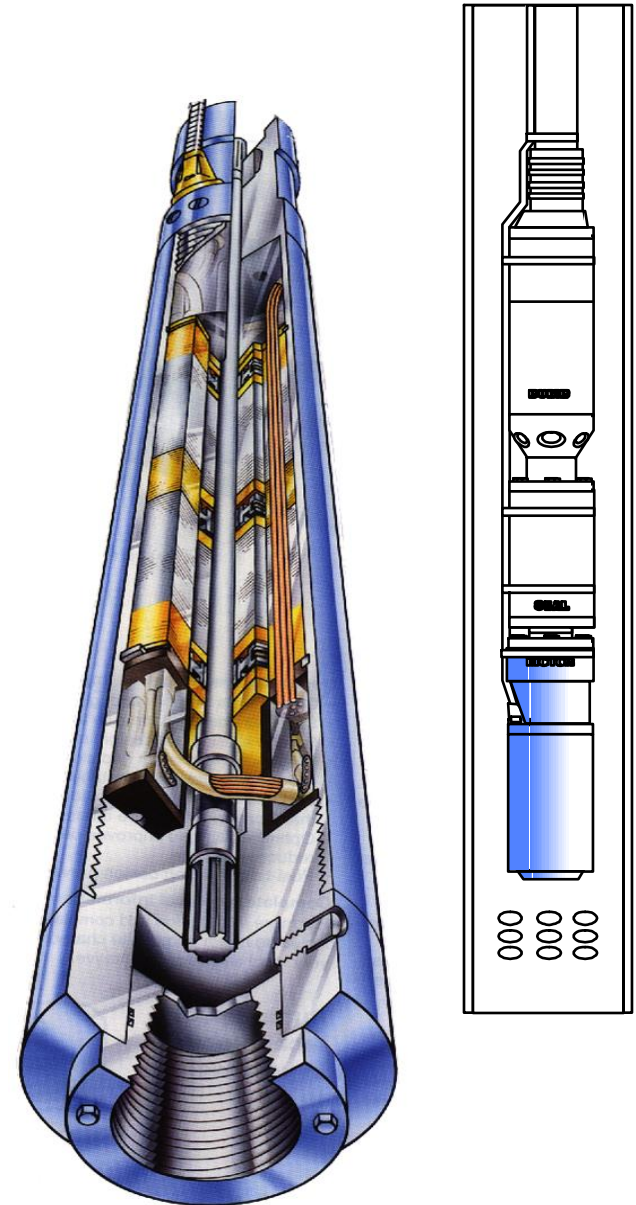


Centrifugal FC650 - Single Stage, 60 Hz, 3600 RPM, 1.0 cP, 1.00 sg

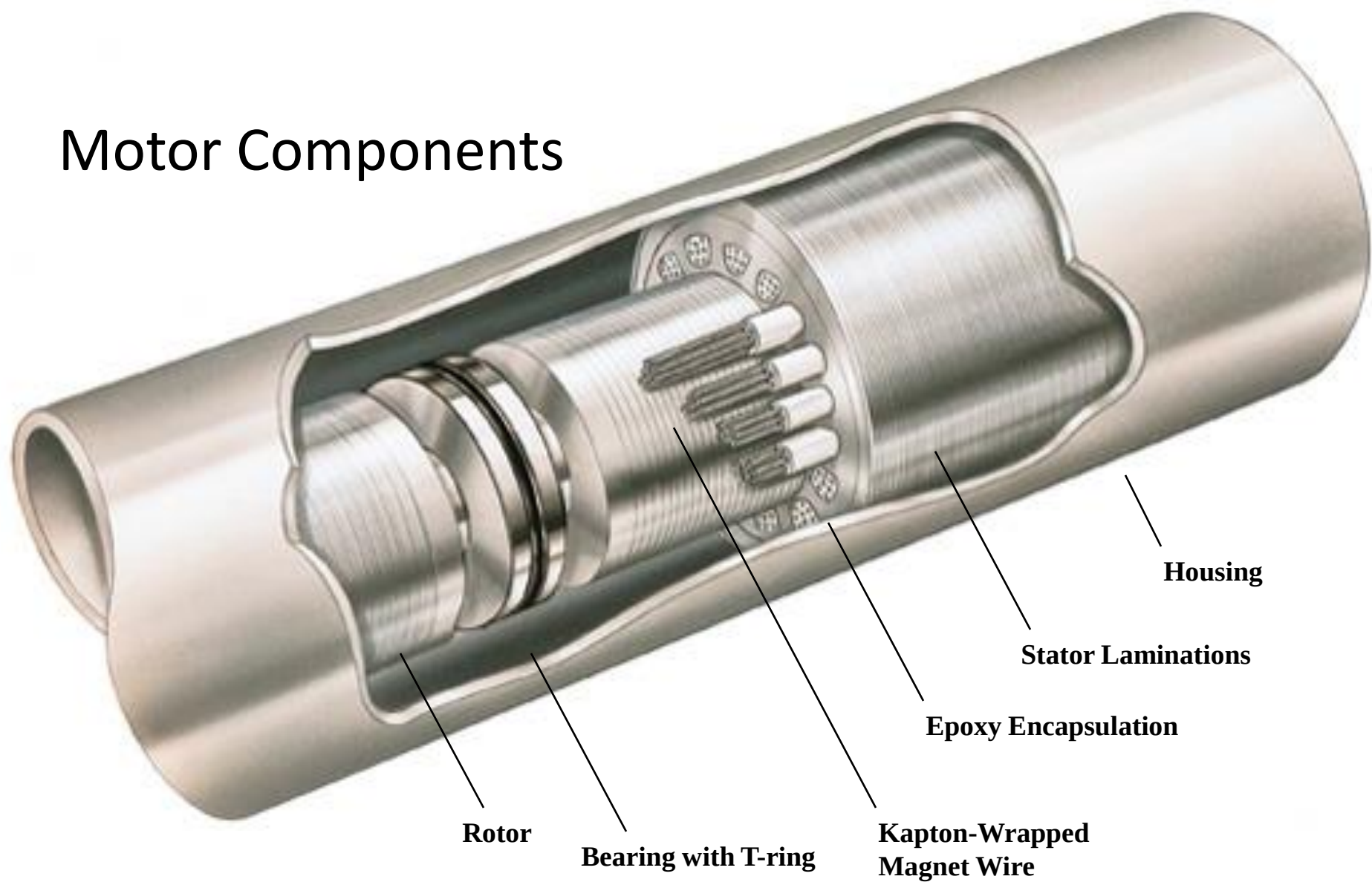


Submersible Motor

- Motor is oil filled
- It is a two-pole, three phase, 3,600 RPM design
- Motor components are designed to withstand 500°F temperatures

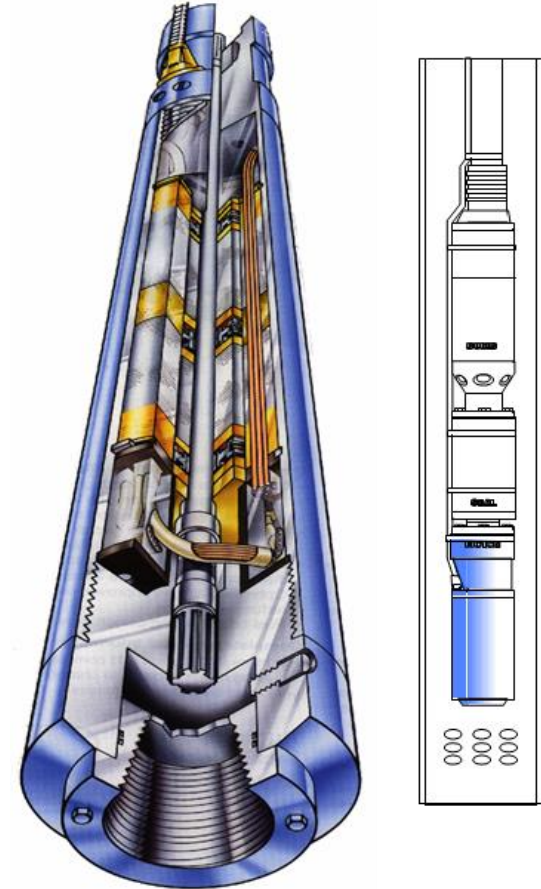


Motor Components



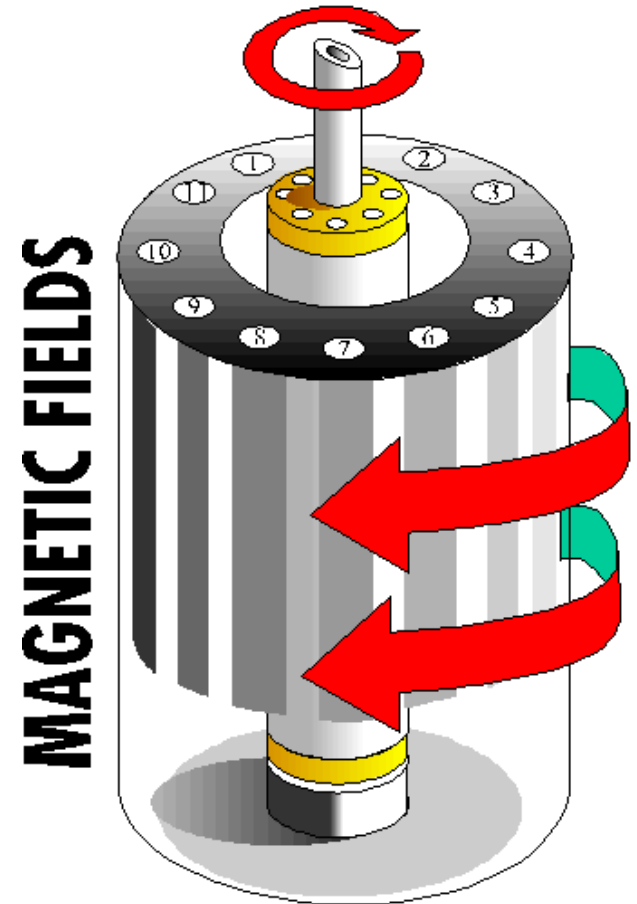
Submersible Motor

- Motor cooling is achieved by:
 - Internal oil circulation
 - Flow of well fluid along outside skin



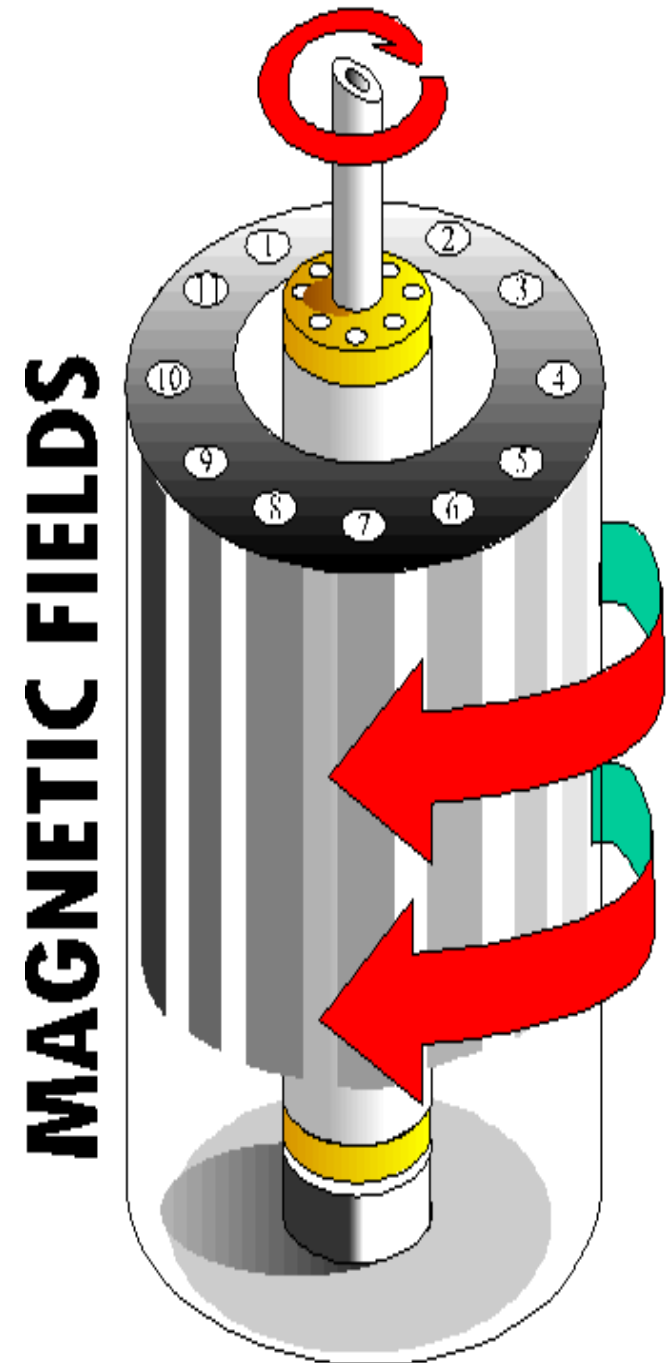
Downhole Motor

- 3-phase alternating current creates revolving magnetic fields in stator
- Magnetic fields cause rotor and shaft inside stator to spin
- Each rotor is then capable of producing X number of horsepower with given voltage



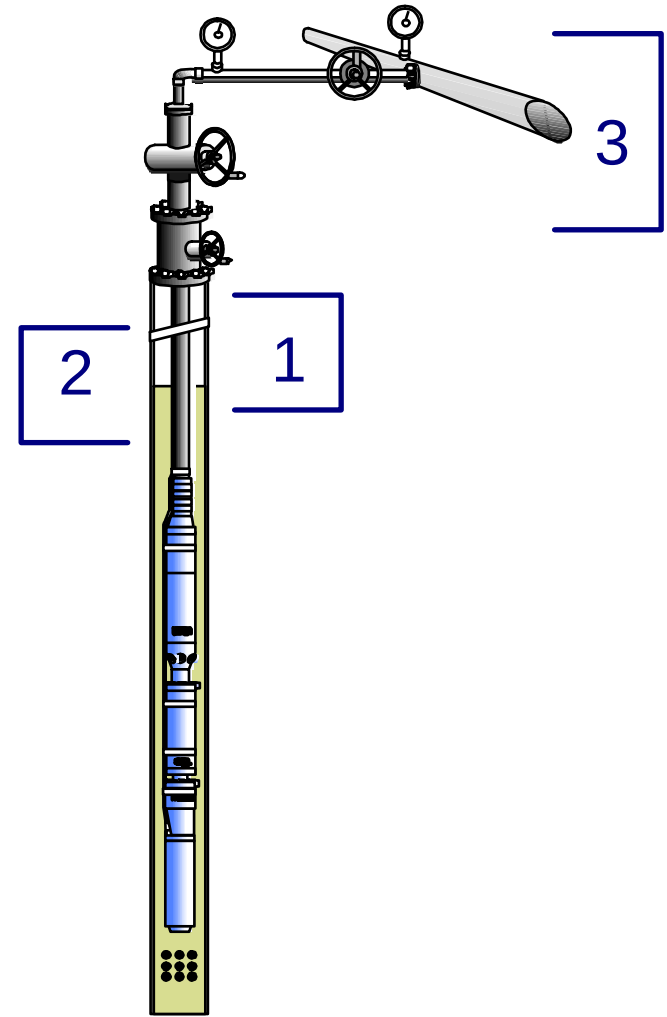
Downhole Motor

- Voltages, on surface, may be changed to adjust horsepower capabilities and performance of motor
- Motor speed = f (frequency)
- Horsepower = f (frequency)



TDH (Total Dynamic Head)

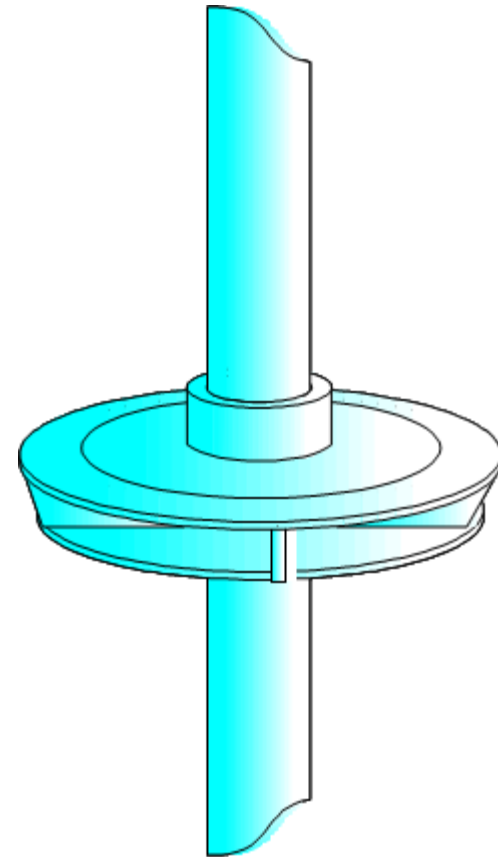
- Pump :
 - 1 Distance from wellbore producing fluid level to surface
 - 2 Friction loss in production tubing
 - 3 Pressure required to overcome flow line back pressure
- $1 + 2 + 3 = \text{TDH}$



Pump Sizing

Example

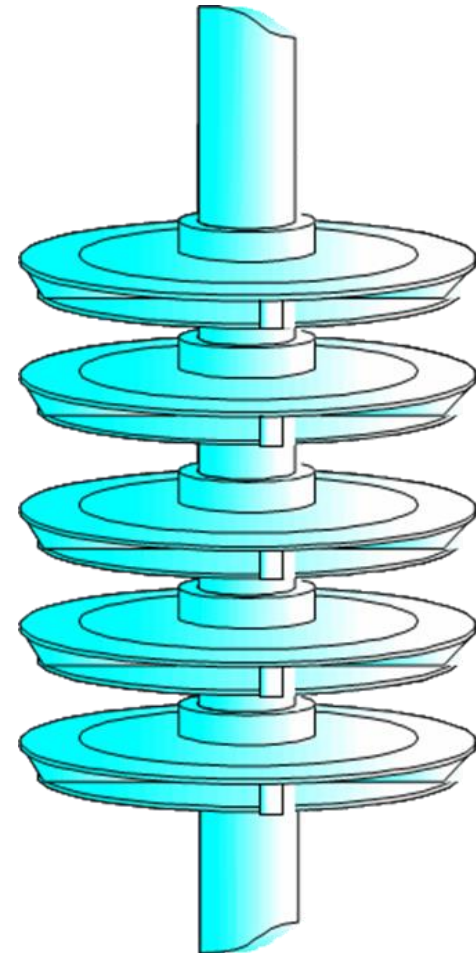
- 1 impeller (stage)
capacity:
 - 10 gallons per minute
 - 10 feet of lift
 - requires 2 hp to run it



Pump Sizing

Example

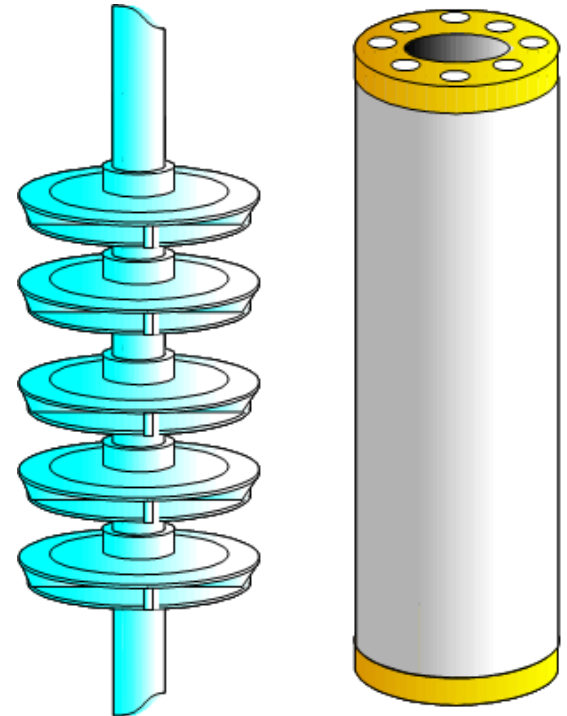
- 5 impeller (stage) capacity:
 - 10 gallons per minute
 - 50 feet of lift
 - requires 10 hp to run it



Motor Sizing

Example -- matching a motor to a pump

- One rotor, 10 HP, drives 5 stages (5 impellers and 5 diffusers)
- If the pump has 50 stages, it needs 10 rotors (100 hp) to run it.



Requirement: 2 hp per stage

Rated: 10 hp per rotor

Downhole Monitoring

- Simple, Reliable Pressure Monitoring
- Pump Intake Pressure (Standard)
 - Well Temperature
 - Motor Winding Temperature
 - Free Water Detection in Motor

Lets discuss; ESP PROBLEMS

- ESP low output
- Vibration
- Sand
- Cavitation
- Cable burnt
- Takes high voltage
- **voltage instability**
- Over heated motor
- High GOR
- Corrosion – erosion – pitting
- Over-loaded motor
- Under loaded motor
- Sludge and asphaltene
- Broken seal
- High water cut

ESP PROBLEMS

Q & A